DexOmni: A Large-Scale Simulation Benchmark for Generalizable Dexterous Manipulation

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Abstract | General-purpose manipulation requires reliable behavior across robot embodiments, objects, and task demands. We introduce DexOmni, a composable simulation data platform and benchmark for cross-embodiment dexterous manipulation. DexOmni supports configurable combinations of five robotic arms and six dexterous hands, together with approximately 200 rigid, articulated, and deformable assets in randomized scenes. The benchmark pairs 18,000 curated human demonstrations with 66 human-verified tasks: Core-50 supports demonstration-based learning and execution tests, while Transfer-16 evaluates asset and relation transfer, skill composition, and deformable manipulation without target-task training demonstrations. Human verification establishes task feasibility, and real-robot experiments validate sim-to-real transfer. Evaluating diffusion, vision-language-action, and world-action policies reveals substantial room for improvement: the best evaluated multi-task policy achieves only 29.1% average success across the 50 Core tasks. Stage-level diagnostics expose recurring failures in establishing contact, coordinating actions, and maintaining achieved goals, with additional sequencing errors in composed tasks. Overall, DexOmni provides a shared experimental foundation for investigating how manipulation capabilities generalize and where reliable physical interaction breaks down.

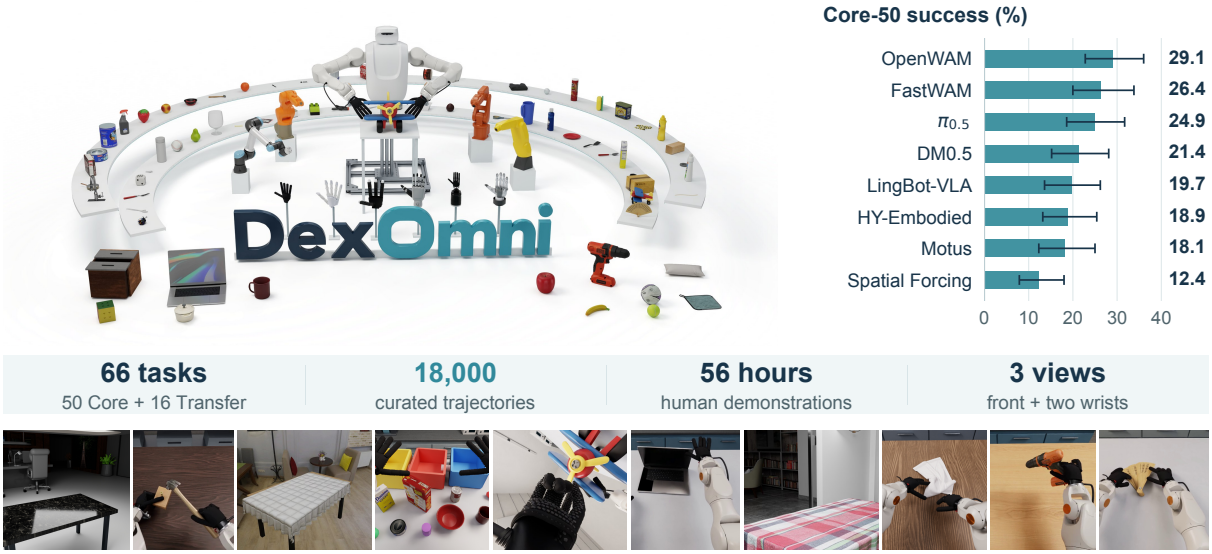


Figure 1 | Overview of DexOmni. Representative assets, environments and interactions accompany 66 tasks and 18,000 curated demonstrations (56 hours; three synchronized views). Right: Core-50 task-macro success of eight multi-task checkpoints; error bars show 95% task-and-episode bootstrap intervals. Poses and cables are illustrative and do not imply control compatibility; the café uses a panoramic background.

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1. Introduction

Dexterous manipulation requires controlling how contacts change during an operation. Moving a mouse must not trigger a premature click; setting two knobs requires both knobs to remain at their target settings simultaneously; and placing a lid requires preserving alignment while releasing it. Evaluating these capabilities requires demonstrations of varied interactions and diagnostics that distinguish initial progress from reliable functional completion. Recent dexterous benchmarks broaden task and embodiment coverage [1–4], but their human-demonstration sets are small and their tasks rarely include ones reserved for evaluation.

DexOmni provides 18,000 curated simulation demonstrations, approximately 56 hours of interaction with three synchronized camera views. This is 5.7–16.4 times the human trajectory count of representative dexterous benchmarks under the convention in Table 1. High-fidelity re-rendering produces varied observations of recorded behavior while preserving state/action supervision. Visual domain randomization and meaning-preserving prompt randomization broaden appearance and language coverage around task-specific configurations. Figure 1 summarizes the asset collection, environment and interaction diversity, demonstration scale, and Core-50 results. Figure 2 connects this data pipeline to evaluation and illustrates correspondence with the physical system.

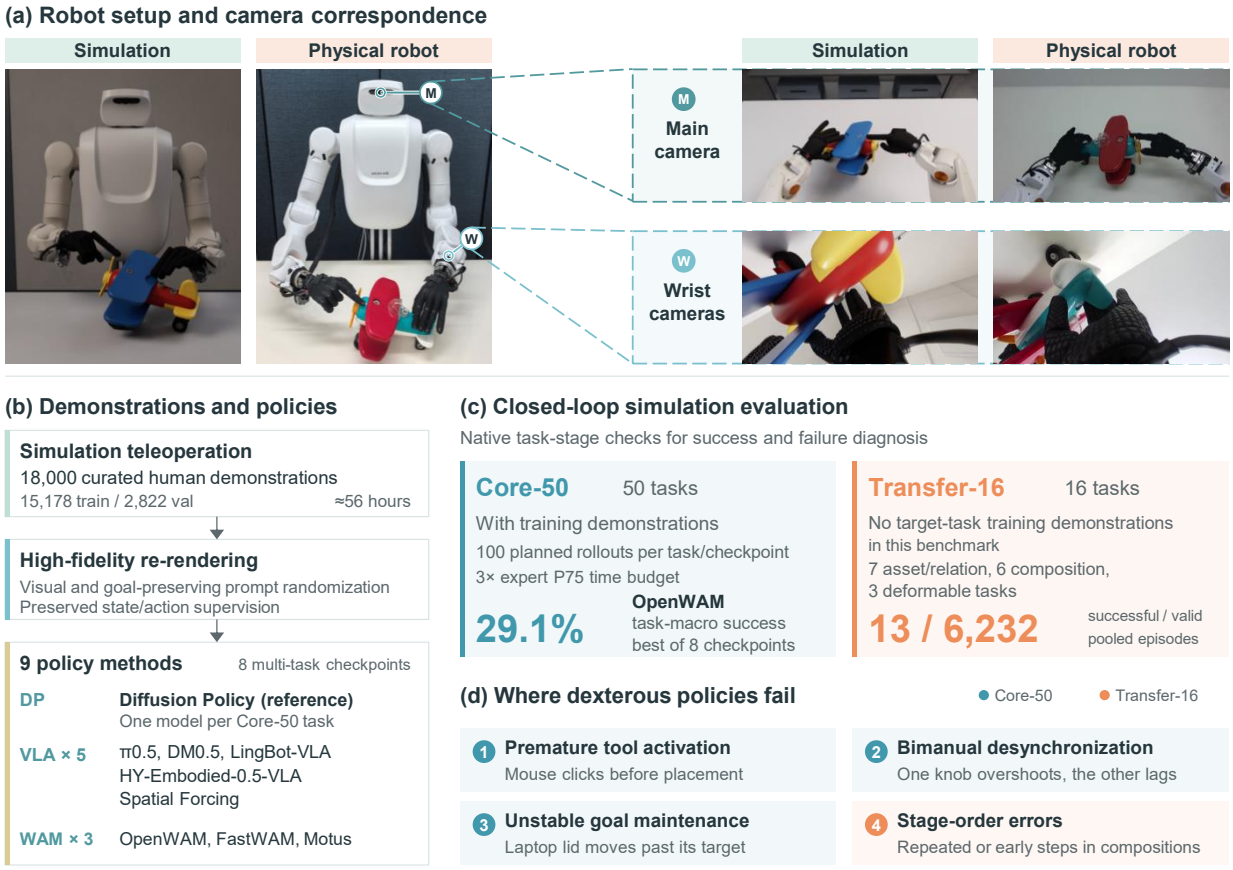


Figure 2 | Platform and evaluation protocol. (a) Simulated and real views of the same toy-aircraft task from the setup camera (not a policy input), wrist cameras (W), and main camera (M); real images show appearance only, and real-robot results are shown in videos only. (b) Data and policies. (c) Evaluation suites. (d) Failure modes analyzed in Section 5.3.3.

The benchmark comprises two complementary suites: **Core-50**, 50 tasks with demonstrations for policy learning and fresh-rollout evaluation, and **Transfer-16**, 16 tasks reserved for generalization evaluation. Core-50 tasks combine functional and numerical understanding, articulation, spatial relations, precise contacts, and bimanual execution (Figure 3). Transfer-16 tests new asset relations and skill combinations, including three deformable interactions without deformable-task demonstrations in this dataset. The simulated embodiment follows a physical platform, and real-robot rollouts of simulation-trained checkpoints are shown in the accompanying videos.

Our Core-50 study compares eight multi-task checkpoints and a single-task diffusion-policy reference. The best multi-task score is 29.1%. Tool use and precision fits are nearly unsolved, and skill-group gaps persist after adjusting for demonstration count and expert duration. Two VLA checkpoints barely condition on which object the instruction names. Failures concentrate where closing the grasp fires a button, where both hands must hold their targets together, and where reached goals must be kept. These findings motivate evaluating what happens after acquisition, alongside aggregate success.

Our contributions are:

1. A large human-demonstration dataset, re-rendered with visual and prompt randomization.
2. Core-50 and Transfer-16: 50 demonstration-backed tasks and 16 evaluation-only asset, skill-composition, and deformable tasks, scored by native stage checkers.
3. A comparison of DP, five VLA, and three WAM policies with a covariate-adjusted skill analysis and a language-conditioning probe, locating failures at three hand transitions and, in compositions, at step order and completion.

2. Related Work

DexJoCo, DexVerse, Bench2Dex, and LabDex cover functional and bimanual manipulation with diverse embodiments [1–4]. DexMimicGen expands human seeds through automated trajectory generation, whereas SPD studies simulation pretraining and real-data adaptation [5, 6]. Table 1 compares data scale, randomization, policy coverage, and evaluation protocols. DexOmni combines these dimensions with three WAM checkpoints and a separate dexterous task-transfer suite.

LIBERO, CALVIN, DexArt, RoboDojo, and RoboTwin motivate transfer, capability evaluation, and randomization [7–11]. Earlier dexterous suites learn mainly by reinforcement or from small demonstration sets [12–14]; large simulated and real manipulation corpora are dominated by grippers [15–19]. Dexterous teleoperation and motion capture [20–22] collect human data outside a benchmark; DexOmni records it in the evaluation simulator. Dexterous sim-to-real learning precedes DexOmni [23, 24]. Section 4.2 describes the dataset counting conventions. Appendix E presents qualitative real-robot rollouts, and Appendix B.5 specifies the proposed matched evaluation protocol.

3. The DexOmni Benchmark

DexOmni has two suites. **Core-50** provides demonstrations for policy learning and fresh-rollout evaluation; **Transfer-16** is reserved for generalization evaluation and contributes no target-task demonstration to the training set. Both suites are scored by native, task-specific success predicates rather than by a shared proxy, and every task was verified in person by the data-collection operators before release (Section 3.2).

Table 1 | Dexterous learning resources (DMG: DexMimicGen; NR: not reported). Human and generated trajectories are counted separately; catalogs and evaluation scopes differ; DexOmni shaded.

	DexJoCo	DexVerse	Bench2Dex	DMG	SPD	**DexOmni**
Task catalog	11	100	26	9	6 scenes	**50 + 16**
Human demos	1,100	3,180	~1,300	60	~2,000	**18,000**
Generated demos	—	—	—	21,000	—	—
Language variation	Digits / words	NR	NR	NR	No input	**32 / 8 per goal**[a]
Test-only tasks	NR	NR	NR	NR	NR	**16 tasks**
Transfer setting	Visual; language	Visual; embodiment	Visual / geometric	Initial states; objects	Real-task finetuning	**Assets; skills; soft objects**
Policy coverage	ACT, DP, VLA	DP, 3D DP, VLA	ACT, DP, VLA	BC-RNN, DP	Diffusion transformer	**DP; 5 VLA; 3 WAM**
Extra diagnostics	Failures; grounding	Category SR	Stages; time; safety; grasp	Generation yield	Stage progress	**Transitions; retention; contact**
Real evaluation	NR	NR	NR	Real demos via sim	Real finetuning	Qualitative[b]

[a]32 train / 8 held-out paraphrases for each of 91 task–goal pairs; separate from Transfer-16. [b]Recorded real-robot rollout videos; no physical success rate. Extra views do not add human demonstrations. Dataset counting conventions: Section 4.2. Comparisons use the cited sources and their reported evaluation scopes.

3.1. Core-50 and Transfer-16

Core-50 tasks combine functional and numerical understanding, articulation, spatial relations, precise contacts, and bimanual execution (Figure 3). Transfer-16 tests new asset relations and skill compositions, including three deformable interactions for which this dataset contains no deformable-task demonstrations. Because the two suites answer different questions, we report them separately throughout: a Core-50 score measures execution on the demonstrated distribution, whereas a Transfer-16 score measures transfer to held-out tasks after fine-tuning on Core-50.

Table 2 | The two DexOmni suites. Core-50 carries demonstrations; Transfer-16 is evaluation-only.

	Core-50	**Transfer-16**
Purpose	Policy learning and execution evaluation	Asset, relation, and skill-composition generalization
Tasks	50	16 (6 EVAL-Comb, 7 EVAL-Gen, 3 EVAL-Flex)
Demonstrations	46–463 training trajectories per task (median 268)	None used for training; 13 tasks use a described duration proxy
Scope	Functional tool use, numerical and spatial grounding, articulation, bimanual manipulation	New asset relations, skill composition, three deformable interactions
Scoring	Task-macro success	Task-macro success and ordered progress

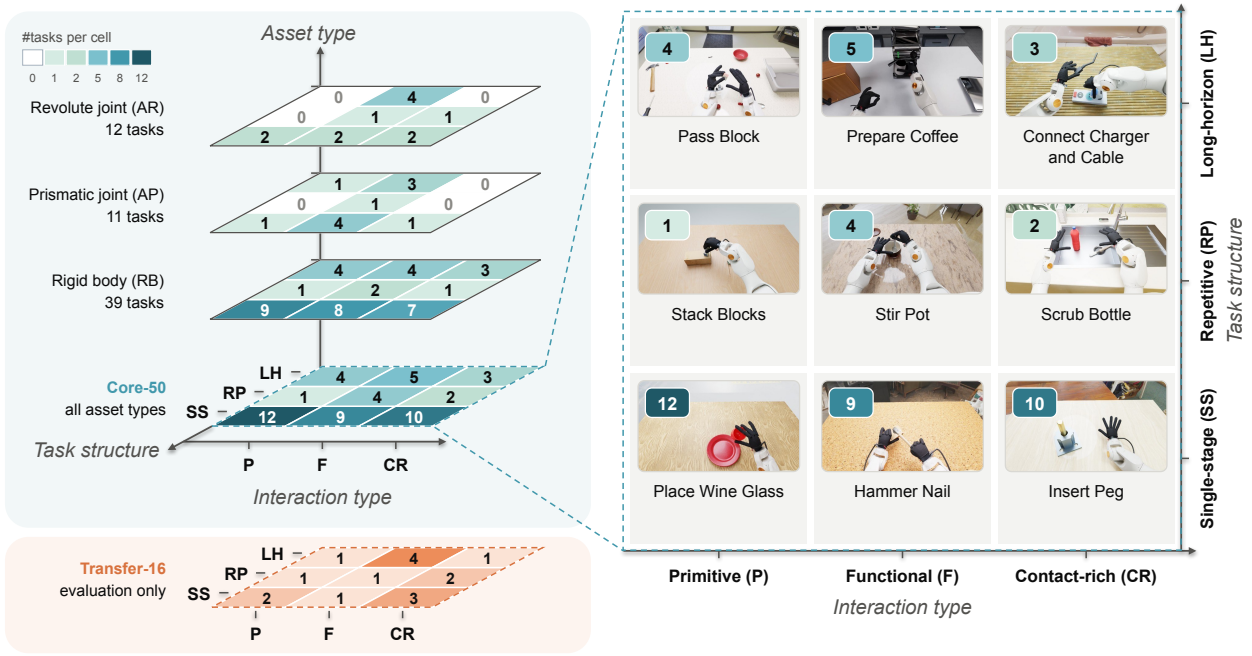


Figure 3 | Three-dimensional task taxonomy of DexOmni. Tasks are classified by interaction type (primitive, functional, contact-rich), task structure (single-stage, repetitive, long-horizon), and asset type (rigid bodies, assets with revolute joints, assets with prismatic joints, deformable objects). A task may have multiple asset-type labels. Each cell gives the number of tasks at the intersection of the indicated categories. The right panel illustrates the nine interaction–structure combinations in Core-50; badges show the corresponding task counts. Evaluation-only Transfer-16 tasks are shown separately in orange, with deformable manipulation reserved for this suite. Table 9 provides the full per-task taxonomy.

3.2. Task Design and Feasibility Verification

Tasks and demonstrations pass a written collection-and-review procedure with separated roles: data-collection operators record and self-check episodes, independent reviewers judge them, and a third person arbitrates disagreements; nobody reviews their own data. Each task has a task card that fixes the goal, the standard initial and final states, completion evidence independent of the program's success flag (object poses, joint states, or images), allowed and forbidden actions with known shortcuts, the time limit, and the environment, task, and checker versions with their code hashes.

A new task or major revision is first piloted with at least 20 operator episodes, all reviewed by hand and covering success, failure, time-out, reset, and boundary states; it enters collection only with no severe defect, at least 98% agreement between labels and the objective evidence, and unambiguous rules. During collection the environment and checker are frozen by hash. A checker change passes reproduction, impact analysis, two-person review, regression tests on normal, failing, boundary, and known-defect cases, and a new fully reviewed pilot; affected episodes are replayed, relabeled, or quarantined. Every recorded episode then passes automatic checks of completeness, version hashes, action plausibility, agreement between the success flag and the final physical state, and duplication. Stratified random blind review covers about 20% of each batch (all of the first batch of a new task or version), and a further 5–10% of passed episodes are blindly re-reviewed. A success flag without objective completion quarantines the task as a suspected checker defect until it is repaired and piloted again.

This procedure matters for how the results should be read. Every Core-50 and Transfer-16 task was verified in person by the data-collection operators in the released environment before evaluation. Each Core-50 task is also completed by its curated demonstrations. For 47 of the 50 Core-50 tasks,

at least one evaluated multi-task checkpoint succeeds in at least one episode. Human verification establishes task feasibility under the verified configurations. The low policy success rates indicate substantial performance gaps under the evaluated protocol (Section 5.3.3). Sources adapted from prior benchmarks, and the modifications made to them, are recorded in Table 10.

3.3. Evaluation Protocol

Simulation uses the fixed native end-effector execution protocol and checkpoint-owned representation, preprocessing, and normalization. Actions are executed at 30 Hz, with each action advancing the simulation by exactly 1/30 s; integration steps are recorded separately. The small source cohort collected with another controller is identified rather than assumed equivalent.

Time budgets. Budgets are fixed before policy evaluation and shared across models. For a Core-50 task i, let $F_{i,(k)}$ denote its demonstration lengths in frames, sorted in ascending order with zero-based index k. The frozen duration table uses the non-interpolated quantile

$$q_i = F_{i,(\min(n_i-1,\lfloor 0.75 n_i \rfloor))}, \qquad H_i = 3q_i/30, \qquad B_i = \lceil 30 H_i \rceil. \tag{1}$$

Here H_i is the budget in simulated physical seconds and B_i the maximum number of executed actions. Source demonstrations are recorded at 30 fps and evaluation advances exactly 1/30 s per action, verified from the physics timestep and substeps. Inference latency, rendering time, and action-chunk length do not redefine this physical budget. The released table is computed from the curated package's source cohorts, including its trajectory validation partition, without using evaluated policy outcomes.

For the 13 non-Flex Transfer-16 tasks no target-task demonstration distribution exists. We instead use $H_i = 3 \sum_{j \in \mathcal{S}_i} q_j/30$, where $\mathcal{S}_i$ is a predeclared mapping to supporting atomic tasks. This is a duration proxy, not the measured P75 of composed trajectories. For example, *Open Laptop and Click Mouse* uses (196 + 125) source frames, giving 32.1 s or 963 actions. The three Flex tasks retain explicit native budgets: 240 s for sponge compression and 180 s each for towel folding and pillow packing. Table 11 lists all Transfer-16 budgets and their mappings.

Stopping and success. After every executed action the native environment checks success, task-defined failure, and timeout. The rollout stops at the first terminal outcome, including within an action chunk; unused predicted actions are discarded. Success requires the complete native predicate, including any release, stability, or dwell requirement. Native failure terminates with zero success; exhausting the budget without success also scores zero. Success reached on the final permitted action is accepted. There is no generic early stop for apparent stagnation or for completing only an intermediate stage. Task-specific irreversible violations retain their documented native rules.

Validity and attribution. An invalid policy output, such as a nonfinite action, is recorded as a policy failure under the declared adapter. Simulator defects, rejected resets, process interruption, and infrastructure failures are recorded separately and do not become numerical policy failures. Wall-clock watchdogs bound infrastructure execution only. Results retain executed steps, physical time, terminal reason, validity, and protocol revision; historical longer-budget successes are never relabeled as successes within a later budget. A repaired environment requires fresh target-task trials, and missing tasks and model/task coverage remain explicit in aggregates.

3.4. Diagnostics and Statistical Reporting

Success and failure alone do not say *where* a policy stops working. We record native-stage diagnostics where supported by the task checker and report their coverage explicitly.

Task success and ordered progress. Primary scores are per-task success and macro-averages over declared task sets. Native executable success predicates and physical-duration holds determine completion. For a task with K_i stages, ordered progress is $P_e = L_e/K_i$, where L_e is the longest valid ordered prefix completed in episode e. Stages are defined by semantic operations; merely approaching or touching an object does not complete a functional stage. Progress never overrides the final success predicate.

Capability-specific diagnostics. Functional completion separates acquiring a tool from operating it. Stability measures include drops, slip, and displacement of an object supported by the other hand when these quantities are observable. Deformable tasks additionally report task-specific geometric progress. Contact-force measures require validated sensors or simulator measurements; command magnitude is never treated as measured force. We distinguish tool acquisition from functional activation, requested numeric setting from any actuator movement, and articulated travel from mere object contact. Capability labels overlap and use disclosed membership; their averages are not added into a new overall score.

Trials and uncertainty. Trials preserve task-local resets, assets, source/goal weights, state/action semantics, and checkpoint-owned preprocessing and normalization. Independent evaluation seeds are shared across compatible model comparisons. Binary success receives Wilson intervals; aggregate uncertainty is reported with the resampling unit specified. Missing or environment-invalid trials are not numerical policy failures. Every aggregate identifies its task and trial coverage, and checkpoint selection, benchmark-development exposure, and historical data exposure are disclosed in the run record.

Reading the diagnostics. Native event-arrival rates and ordered completion scores are different quantities. A conditional transition rate requires joint, order-consistent events, not division of unrelated stage marginals. Retention diagnostics identify both the achieved condition and the subsequent observation interval; success-triggered termination can give episodes unequal exposure. Recorded action commands are not measured wrist motion or contact force.

4. The DexOmni Data Platform

DexOmni pairs a teleoperated demonstration corpus with a re-rendering pipeline. Collection produces exact state and action supervision at the cost of a fixed set of recorded appearances; re-rendering then varies the observations *around* that recorded behaviour while the interaction and its labels stay fixed. The two halves are described in turn.

4.1. Collection System and Operators

Demonstrations are collected in NVIDIA Isaac Sim 5.1 [25]. MANUS Prime I gloves capture finger articulation, and wrist-mounted HTC VIVE Trackers provide global wrist poses [26, 27]. After frame registration and temporal alignment, wrist motion and a palm-centred 21-keypoint hand representation are converted to commands. The arm branch produces feasible end-effector targets through robot kinematics; Wuji retargeting maps hand keypoints to joint targets, with 20 outputs for

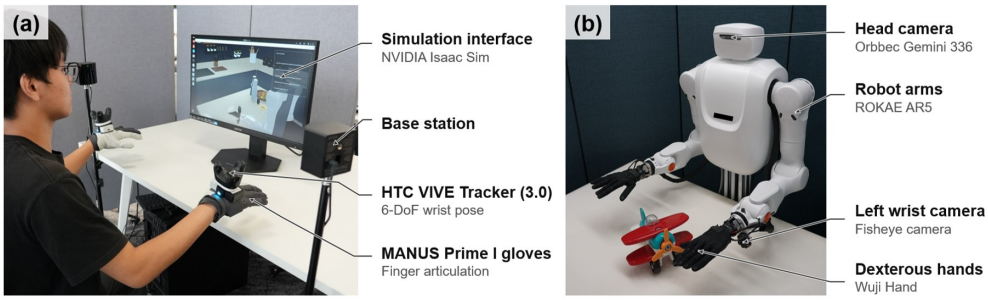


Figure 4 | Human demonstration collection. Gloves record finger articulation and wrist trackers record hand motion. Retargeted commands drive the simulated arm–hand system while the operator observes the scene.

the first-generation Wuji Hand [28]. Simulation records synchronized observations, measured state, commanded actions, language, and source metadata (Figure 4).

At least ten operators contributed over 1,050 collection sessions, which is why the feasibility procedure of Section 3.2 separates recording from review. All demonstrations are simulated: no physical robot recordings enter the training set, and the physical platform of Section 4.5 is used only for the qualitative rollouts of Appendix E.

4.2. Dataset Composition and Identity

The released dataset contains 18,000 curated trajectories: 15,178 training and 2,822 validation. Selection uses task-group quotas of 500, 300, or 100 trajectories, favouring shorter eligible demonstrations within each quota. This is a curated distribution, not an unbiased sample of all collected behaviour. Three stages of one multi-stage source task share a quota, giving 52 quota groups. Three simultaneous camera views do not multiply the interaction duration. Across the package the median trajectory has 262 steps (10th–90th percentile 96–705); per Core-50 task, training trajectories range from 46 to 463 (median 268) and expert P75 durations from 2.5 to 49.2 s (median 12.7 s). Split membership is defined at trajectory or session level and re-rendered derivatives inherit it; normalization fitted for this package uses training data only, and a checkpoint that inherits another normalization or has historical exposure retains that provenance.

Table 3 lists the counting units that recur through the paper. Keeping them separate matters: the 54,000 video streams are three views of the same 18,000 interactions, and the raw collection before deduplication and curation is not the training size of any checkpoint.

Table 3 | Distinct counting units in the dataset and catalog. The video count is three views of the same interactions, not three times the data.

Unit	Count
Curated demonstration trajectories	18,000
Training / validation trajectories	15,178 / 2,822
Recorded time steps	6,053,555
Camera videos (three per trajectory)	54,000
Native task identities / quota groups	54 / 52
Core-50 / Transfer-16 tasks	50 / 16

The evaluation catalog contains the Core-50 tasks and excludes one source task, Chinese Chess, for every model. This is a scope exclusion: it does not remove demonstrations from the 18,000-trajectory

package and does not rewrite any checkpoint's training provenance. A mapping record links each paper task to its source task or cohort, goal, asset identities, episodes, and executable evaluation task.

4.3. High-Fidelity Re-rendering

Re-rendering varies appearance while replaying the recorded interaction. Final-package capture records use 32 samples per pixel; higher-sampling presentation renders are identified separately. The implemented bank includes four table materials, three background configurations, and three ceiling styles, with texture scale and orientation, surface roughness and tint, and illumination direction, intensity, and colour-temperature variations. One verified source camera configuration samples each camera once per trajectory within ±2 mm translation and ±0.3° rotation per axis, retaining intrinsics. Episode records identify source trajectory and runtime, render revision, appearance seed and configuration, and camera configuration. Appearance is sampled per trajectory and remains fixed during replay; independently resampling appearance every frame is outside this static-domain protocol, and any temporal sensor noise would require a separate model.

Not every factor can be varied by replay alone. Table 4 states what each factor must preserve. Background and appearance changes can be applied during replay because the recorded actions remain valid; geometry and dynamics changes alter the reachable interaction and require valid re-execution rather than frozen-state replay. We use the taxonomy of Table 4 as the rule for adding variation: a factor is admitted only with the evidence its row requires.

Table 4 | Visual, language, and physical variation and the labels each must preserve. Geometry and dynamics changes require evidence beyond visual state replay.

Factor	Examples	Must preserve
Appearance	Material, lighting, background, ceiling	Preserve recorded interaction and task-relevant visual semantics.
Camera	Extrinsic perturbation about calibration	Preserve timing; transform camera-frame labels if affected.
Language	Canonical instruction and paraphrases	Preserve object, numerical goal, direction, order, and constraints.
Task geometry	Object pose, goal relation, native reset	Retain each source task's joint sampling distribution and compatible actions.
Dynamics	Mass, friction, compliance, actuation	Requires valid re-execution; not established by frozen-state re-rendering.

Camera perturbations are composed with task-specific nominal calibration. Within Core-50 evaluation, which of the four table materials, three colour palettes, or two roughness levels is drawn does not measurably change success (Section 5.3.4). Those factors lie inside the training distribution, so this is an invariance check rather than a visual-generalization result.

4.4. Language Bank and Prompt Sampling

The frozen language bank contains 108 task–instruction entries, of which 91 occur in the final 18,000-trajectory manifest. For these 91 active entries, there are 32 training and eight held-out paraphrases per entry, yielding 2,912 training and 728 held-out paraphrases. For example, "Pick the apple and place it into the bucket" and "Grab the apple and put it in the bucket" preserve the same goal. Replacing apple with banana, changing a knob angle, or changing operation order instead requires an episode with the corresponding executable goal, which is why those variations are task variations rather than paraphrases. Held-out strings have no exact overlap with any active canonical or training

string; this is a lexical disjointness check, and semantic equivalence additionally requires preserving the episode's target and constraints. Inactive bank entries are not counted as demonstrated goals.

The packaged sampler hashes the language seed, epoch, episode identity, frame index, draw index, and split to choose an instruction, and offers training, canonical, and unseen modes. The reference training mixture is 20% canonical and 80% training paraphrases; fixed validation uses the run's declared language mode. Language sampling does not resample the scene, cameras, or physics. A model record must identify the language bank revision, the actual sampling policy, prompt preprocessing, and whether conditioning is text, a structured goal, or a fixed task identifier. Section 5.3.2 examines whether policies condition their actions on the object specified in the instruction.

4.5. Physical Platform and Simulated Correspondence

The platform uses ROKAE AR5-08 arms (seven degrees of freedom, 5 kg rated payload, 683 mm reach) and first-generation Wuji Hands with 20 actively controlled joints per hand [29, 30]. The simulated kinematics, asset scale, cameras, state and action frames, and timing follow this physical system, and the correspondence record is given in Appendix B.1.

4.6. Portability Across Arms, Hands, and Physics Backends

DexOmni is organized around a task and asset layer that is independent of the particular hardware that executes it, so that a task definition, its assets, its success predicate, and its demonstration set survive a change of arm, hand, or physics back-end. The released code supports the substitution of several commercial arms, including the ROKAE arms used in the reported experiments and additional supported arms from KUKA, UR, ABB, and FANUC; several dexterous hands, including the first- and second-generation Wuji Hands and the Sharpa Hand; and two physics back-ends, NVIDIA PhysX and Newton. The benchmark therefore measures a manipulation capability rather than a specific piece of hardware: the same Core-50 and Transfer-16 definitions can be instantiated against any supported arm–hand pair, and the demonstrations can be re-collected on that pair without editing the task itself.

All experiments reported in this paper use one configuration: the ROKAE AR5-08 arms and first-generation Wuji Hands of Section 4.5, with NVIDIA PhysX as the physics back-end. This is the configuration of the 18,000-trajectory corpus, of every collection session, and of every Core-50 and Transfer-16 result in Section 5. The other arms, hands, and back-ends are supported by the platform and are the natural basis for studying transfer across embodiments, but they do not contribute episodes to the results reported here, and the numbers in this paper should not be read as performance on those configurations.

5. Experiments and Analysis

We evaluate eight multi-task checkpoints and a single-task diffusion-policy reference on Core-50. Section 5.1 describes the experimental setup. Section 5.2 presents the main results on Core-50 and Transfer-16, and Section 5.3 examines demonstration scale and task duration, instruction conditioning, failure modes, task grouping, and evaluation reliability.

5.1. Models and Experimental Setup

We train eight multi-task checkpoints on the Core-50 demonstrations. The five vision-language-action (VLA) models are $\pi_{0.5}$ [31], DM0.5 of the DM0 family [32], LingBot-VLA [33], HY-Embodied-0.5-VLA [34], and Spatial Forcing [35], which aligns the visual features of $\pi_{0.5}$ with a pretrained

3D foundation model during training. The three world-action models (WAMs), OpenWAM [36], FastWAM [37], and Motus [38], combine action prediction with modeling of future observations. Each checkpoint executes its integration's default action chunk per query (Table 5, Steps/plan). The selected checkpoints are $\pi_{0.5}$ at 120,000 updates, DM0.5 at 86,000, OpenWAM at 70,000, Motus and FastWAM at 100,000, Spatial Forcing at 90,000, HY-Embodied at 125,000, and LingBot-VLA at 151,000; they are fixed, not chosen by benchmark success. The single-task reference, Diffusion Policy (DP) [39], trains one 30,000-update model per Core-50 task. Without a predeclared routing protocol it is not run on Transfer-16; this is not a measured zero-shot failure. Pretraining data and compute differ across models, so the comparison does not isolate architecture.

5.2. Benchmark Results

5.2.1. Performance on Core-50

Table 5 reports all 50 Core-50 tasks for the eight multi-task checkpoints (per task in Appendix C). For analysis we group the tasks by the hand skill they demand (Section 5.3.4; Figure 5a).

Table 5 | Core-50 task-macro success (%), 100 episodes per task and checkpoint; 95% intervals in Table 13. Steps/plan: control steps (30 Hz) per policy query. Tasks ≥50%: number of the 50 tasks with at least 50% success. P/F/CR: primitive (17), functional (18), contact-rich (15) tasks. @1×: success within the expert p75 duration. Early: episodes ended by an irrecoverable failure. Stall: the percentage of instrumented time-outs with no new checker state after the expert p75 duration (Section 5.3.3). Shaded: best (lowest for Early, Stall).

Model	Steps/plan	SR	Tasks ≥50%	P	F	CR	@1×	Early	Stall
OpenWAM	16	**29.1**	**11**	40.8	**23.8**	**22.1**	8.3	**10.9**	**65**
FastWAM	16	26.4	9	**42.7**	20.2	15.3	10.2	13.0	72
$\pi_{0.5}$	30	24.9	10	40.4	14.6	19.7	**14.6**	14.1	78
DM0.5	30	21.4	9	37.7	10.0	16.7	10.9	13.9	80
LingBot-VLA	30	19.7	8	26.9	14.3	17.9	12.0	12.7	83
HY-Embodied	30	18.9	8	27.1	13.3	16.4	11.0	14.4	81
Motus	16	18.1	5	30.1	11.5	12.3	5.4	15.6	70
Spatial Forcing	16	12.4	3	17.9	8.5	10.9	5.6	16.5	75
DP (single-task)[†]	—	9.7	3	reference; 9.9 on 11 matched tasks					

[†]One 30k-step model/task. Earlier goal sampler: 38 tasks; matched: ten retrained single-arm tasks and Open Latched Door ($\pi_{0.5}$: 29.1); separate: Lift Carton. All 50 contribute to SR. HY-Embodied: 99 valid episodes on Place Wine Glass.

Policies move objects and mechanisms; tools and precision fits remain unsolved. OpenWAM has the highest success, 29.1%, and picking the best checkpoint per task would give 35.4%. Success is ordered by what the task asks of the fingers beyond closing them. Pushing or flicking without a grasp (two tasks, 51% pooled over checkpoints), moving a hinge, drawer, lever, or lid (43%), and grasp–transport–release including handovers and object selection (35%) are the relatively better-performing groups in the suite; all nine tasks with mean success above 40% are of these kinds. Operating a trigger, button, knob, slider, or shutter with the fingers drops to 14%, pouring to 13%, and regrasping or reorienting an object in or between the hands to 11%. Two groups are essentially unsolved: acting on a third object through a held tool (137 of 4,800 episodes, 2.9%; hammer, spoon, set square, and spatula together 9 of 3,200) and precision fits such as peg insertion, stacking, or face alignment (107 of 4,800, 2.2%). The best checkpoint on any of these twelve tasks reaches 21% (OpenWAM on Stir Pot); on ten of them none exceeds 14%. Skill-group gaps persist after

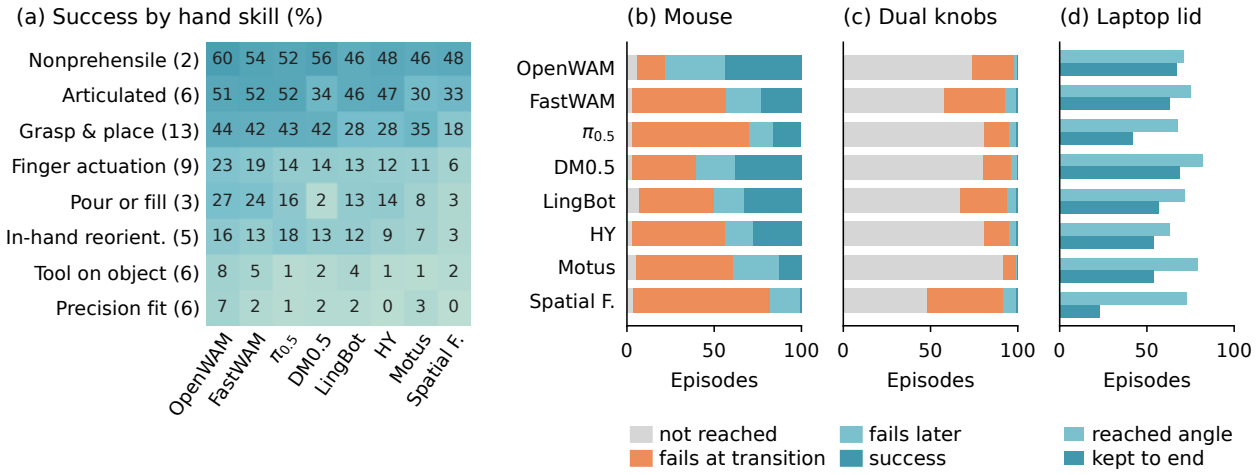


Figure 5 | Core-50 diagnostics, eight multi-task checkpoints. (a) Task-macro success by hand-skill group (number of tasks in parentheses). (b)–(d) Episodes at the three transitions of Section 5.3.3; grey: transition never reached, orange: failed at it, light blue: failed after it. (b) Mouse moved 8 cm onto its pad before any click. (c) Both knobs within 5° of target at once (grey: at most one knob). (d) Laptop display ever at the requested angle, and still there at the end.

adjusting for demonstration count and expert duration: precision fits have more demonstrations per task (median 413) than grasp–transport–release (392), and with log demonstration count and log expert duration as covariates the tool, precision, in-hand, and finger-actuation groups stay 20–25 points below grasp–transport–release, with task-bootstrap intervals excluding zero (Section 5.3.1).

Checkpoints differ in using a grasped object, not in grasping it. On grasp–transport–release tasks four checkpoints are within two points (OpenWAM 44, $\pi_{0.5}$ 43, FastWAM 42, DM0.5 42), and the OpenWAM–$\pi_{0.5}$ gap on primitive tasks is +0.5 points (paired task bootstrap 95% interval −6.9 to +7.6). The same gap on functional tasks is +9.2 (+4.7 to +13.9): Pour Seasonings 60 versus 34%, Move and Click Mouse 44 versus 16, Press Soap Pump 33 versus 9, Stir Pot 21 versus 4. Section 5.3.3 locates this gap at specific transitions: OpenWAM clicks before the mouse is ready in 16 of 100 episodes and $\pi_{0.5}$ in 67, OpenWAM keeps the laptop display at the requested angle in 67 and $\pi_{0.5}$ in 42, and OpenWAM completes the handover to the receiving hand in 41 and $\pi_{0.5}$ in 20. Two VLA checkpoints place fixed objects as well as the leaders but collapse when the instruction selects the object: HY-Embodied and LingBot-VLA score 55 and 61% on Place Cup on Plate ($\pi_{0.5}$ 62) but at most 22% on Place Ball in Bowl, Place Fruit in Bucket, and Place Largest Fitting Ball, where $\pi_{0.5}$ scores 35–75%. They barely read the object name: in an offline probe (Section 5.3.2), naming the other ball moves the 1-s palm endpoint of $\pi_{0.5}$ and DM0.5 by twice the resampling change (15 and 17 versus 7 and 8 cm) but that of HY-Embodied and LingBot-VLA by no more (7–9 cm), and with the swapped name they still head for the original ball in 57 and 71% of first frames (DM0.5 10%). Motus is a strong picker (76% on Place Ball in Bowl) but near zero wherever a mechanism must be held open or a load carried (Open Micropipette Box 8%, Open Drawers 4%, Lift Carton 8%). Spatial Forcing is weakest in every group and, unlike all other checkpoints, worse on single-arm (3.8%) than on bimanual tasks (14.5%).

World-action models finish later but finish more. The three WAM checkpoints complete their successes at a median of 1.13–1.27 p75 durations versus 0.88–0.99 for the four 30-step VLA checkpoints, and more of their time-outs are still making progress after 1× (14.5–18.7% of episodes versus

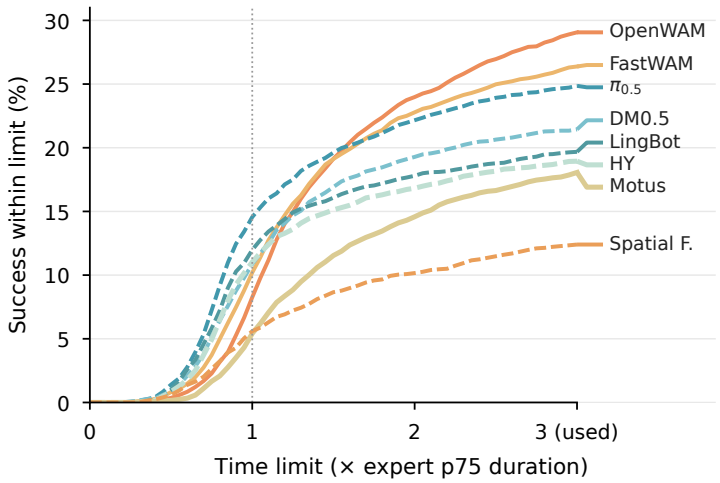


Figure 6 | Task-macro success counting only episodes completed within a given multiple of the expert 75th-percentile duration. Shorter limits are prefixes of the evaluated 3× rollouts, not reruns, so the curves do not estimate what a longer limit would add. Solid: world-action models; dashed: VLA checkpoints. Success at 1×/1.5×/2×/3× is OpenWAM 8.3/19.1/24.0/29.1; FastWAM 10.2/19.1/22.8/26.4; $\pi_{0.5}$ 14.6/19.6/22.2/24.9; DM0.5 10.9/16.9/19.3/21.4; LingBot-VLA 12.0/16.1/17.8/19.7; HY-Embodied 11.0/15.1/16.8/18.9; Motus 5.4/11.5/14.6/18.1; Spatial Forcing 5.6/8.7/10.2/12.4. Twelve of the 28 checkpoint pairs change order between 1× and 3×.

10.1–11.8%). Counting only completions within 1× (Table 5, @1×), $\pi_{0.5}$ leads with 14.6% and OpenWAM ranks sixth with 8.3%; 12 of the 28 checkpoint pairs change order between the two limits (Figure 6). The WAMs execute 16 control steps per query, but Spatial Forcing, a VLA that also executes 16, is not slow (median 1.09), so the query cadence alone does not explain the difference; @1× is therefore the secondary score (Section 3.3).

Reliability and feasibility. The 95% interval over tasks and episodes (22.8–36.0% for OpenWAM, Table 13) mainly reflects the task mix (episodes alone: 28.0–30.1%). Paired task resampling orders $\pi_{0.5}$ above DM0.5 in 99% of draws but HY-Embodied above Motus in only 61% (Section 5.3.4). Leaving out any one task swaps at most HY-Embodied and Motus and moves no score by more than 1.4 points. Every task is completed by verified human demonstrations (Section 3.2). For 47 of the 50 Core-50 tasks, at least one evaluated multi-task checkpoint succeeds in at least one episode.

Single-task reference. DP averages 9.7% over 50 tasks: 38 use an earlier goal sampler, 11 use the current protocol, and Lift Carton is evaluated separately (Appendix C). On the 11 matched tasks it reaches 9.9%, against 22.7–34.5% for seven multi-task checkpoints and 9.3% for Spatial Forcing. This does not separate multi-task data sharing from pretraining or model scale.

5.2.2. Generalization to Transfer-16

Table 6 groups Transfer-16 results by asset/relation transfer, skill composition, and deformable manipulation, with ordered progress crediting the longest completed prefix of stages. Transfer-16 is much harder than Core-50, and we rank checkpoints by ordered progress, which also credits partial completion. OpenWAM's 12.9% exceeds HY-Embodied, LingBot-VLA, and Spatial Forcing by 6.0, 3.7, and 2.6 points (paired intervals exclude zero); the other four are within 1.7 points.

Table 6 | Transfer-16 results (%): task-macro success (SR) and ordered progress (Prog.) within the declared time limit over the 7 Gen, 6 Comb, and 3 Flex tasks; 50 episodes per task and checkpoint (Stack Boxes 37 and Spray Shoe 43, whose missing resets fail for every checkpoint; at most two invalid episodes elsewhere). Gen02 logs no graded hinge progress and is excluded from Prog. DP is not run (Section 5.1). Shaded: best per column.

Model	Asset/relation		Skill composition		Deformable		All 16	
	SR	Prog.	SR	Prog.	SR	Prog.	SR	Prog.
$\pi_{0.5}$	0.0	14.0	0.3	11.7	0.0	8.3	0.1	11.9
DM0.5	**1.4**	9.2	**0.7**	**14.4**	0.0	8.8	**0.9**	11.2
LingBot-VLA	0.0	13.3	0.0	4.2	0.0	11.3	0.0	9.3
HY-Embodied	0.0	9.6	0.0	4.6	0.0	6.2	0.0	6.9
OpenWAM	0.9	14.8	0.0	12.5	0.0	10.2	0.4	**12.9**
Motus	0.3	**15.8**	0.0	9.5	0.0	**11.7**	0.1	12.4
FastWAM	0.0	12.5	0.0	11.3	0.0	11.3	0.0	11.8
Spatial Forcing	0.3	12.7	0.0	7.9	0.0	10.5	0.1	10.3

Human verification establishes the feasibility of Transfer-16 tasks under the verified configurations (Section 3.2). The low success rates indicate substantial policy performance gaps under the evaluated protocol. Success is close to zero on all three families, which makes aggregate success a weak discriminator: ordered progress separates the checkpoints where success cannot. The three deformable tasks are unsolved by every checkpoint, and the compositions, which require completed skills to be chained, are the hardest family after them (Section 5.3.3).

5.3. Detailed Analysis

5.3.1. Does Demonstration Scale Explain the Remaining Gap?

We finetune $\pi_{0.5}$ separately on *Place Ball in Bowl* (Task 24) and *Squeeze Scissors* (Task 43) on nested subsets of 10, 50, 100, 200, and all 438 and 439 training trajectories, each run from the same pretrained initialization for 20,000 updates at batch size 32, with one training seed and the same 100 evaluation seeds, goals, and prompts per task (Appendix D).

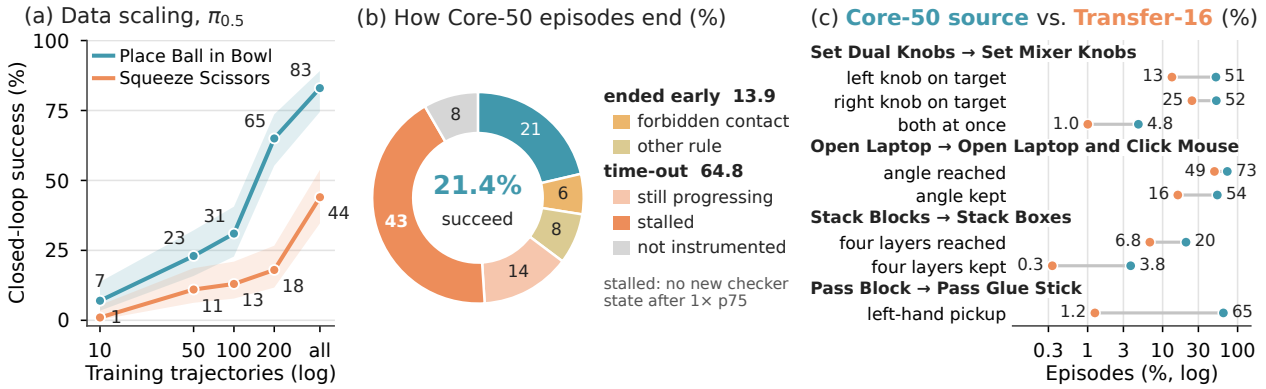


Figure 7 | (a) Single-task data scaling for $\pi_{0.5}$, 20,000 updates per run (all: 438 and 439 trajectories); each point is one checkpoint on 100 closed-loop trials, with 95% Wilson bands over rollouts, not over training seeds or subsets. (b) How Core-50 episodes end, task-macro, mean of the eight multi-task checkpoints (each in Figure 8). (c) Matched hand transitions in Core-50 sources and Transfer-16, pooled over the eight checkpoints (800 source and 296–400 transfer episodes per task).

Success rises at every tested size (Figure 7a): Task 24 from 7% to 83% and Task 43 from 1% to 44%,

and going from 200 trajectories to the full pool still adds 18 and 26 points. The functional scissors task remains much harder. These task-local curves do not establish a cross-task scaling law or isolate an architectural cause, and normalization uses the full training pool at every size.

We also examine whether the Core-50 skill-group gaps persist after adjusting for demonstration count and expert duration. The hand-skill groups differ in how many training trajectories their tasks have and how long the expert takes (Table 7), so group differences could reflect data volume or episode length rather than the skill. Across the 50 tasks, mean success over the eight checkpoints has Spearman $\rho = 0.27$ with the number of training trajectories and $\rho = -0.71$ with the expert P75 duration. Regressing task success on the group indicators alone explains $R^2 = 0.62$ of its variance; the two covariates alone explain 0.37. With both covariates added ($R^2 = 0.73$), duration remains negatively associated with success, whereas the interval for demonstration count includes zero. The finger-actuation, in-hand, tool, and precision groups remain 20–25 points below grasp–transport–release. The in-hand and tool groups have few demonstrations, but precision fits have a similar median demonstration count to the higher-performing grasp–transport–release group. By quota, the seven 100-trajectory tasks average 6.1%, the 24 300-trajectory tasks 23.4%, and the 19 500-trajectory tasks 24.4%; within the 500 quota, the four precision fits average 2.8% and the eight grasp–transport–release tasks 37.8%.

Table 7 | Task success (mean of the eight multi-task checkpoints, %) regressed on hand-skill group over the 50 Core-50 tasks, with grasp–transport–release (GTR) as the reference; the GTR row gives the intercept of the raw model. Covariates are natural logs of the task's training trajectories and expert P75 duration. Brackets: 95% intervals from 4,000 task bootstrap draws stratified by group. Demos and P75: group medians over tasks.

Group	Demos	P75 (s)	Raw	With covariates
GTR (reference)	392	8.2	34.9 [26.9, 43.1]	—
Nonprehensile	362	10.5	+16.4 [−19.3, 52.2]	+18.7 [−16.1, 52.9]
Articulated part	259	6.7	+8.0 [−4.6, 21.9]	+7.4 [−3.0, 19.8]
Finger actuation	272	12.9	−20.9 [−31.2, −10.8]	−19.9 [−27.8, −12.2]
Pour or fill	208	17.7	−21.5 [−34.6, −6.7]	−13.8 [−24.8, −3.3]
In-hand reorientation	89	12.3	−23.5 [−35.2, −11.2]	−22.1 [−31.6, −11.7]
Tool on object	159	23.5	−32.0 [−40.7, −23.5]	−21.3 [−29.9, −11.8]
Precision fit	413	14.7	−32.7 [−41.0, −24.4]	−24.6 [−33.1, −16.6]
log training trajectories			—	−0.8 [−5.1, 5.4]
log P75 duration			—	−11.9 [−16.2, −7.6]
R^2			0.62	0.73

5.3.2. Do Policies Read the Instruction?

To test whether a checkpoint reads the object name, we query it offline on 500 demonstration episodes of Place Ball in Bowl (training and validation trajectories) at frames 0 and 15, 1,000 inputs in total. For each input we sample twice with the true instruction and once with the instruction naming the other ball, and compare the palm position commanded 30 steps (1 s) ahead. The probe uses checkpoints of the evaluated training runs, within 6,000 updates of the evaluated ones. For $\pi_{0.5}$, DM0.5, HY-Embodied, and LingBot-VLA, respectively, swapping the object name shifts the predicted palm endpoint by 14.8, 17.0, 7.3, and 8.5 cm, versus 7.0, 8.2, 8.8, and 8.1 cm between two correct-instruction samples. In the same model order, the correct-instruction endpoint is closer to the demonstration than the swapped one in 92%, 88%, 61%, and 71% of inputs.

For a target-level test we use the $\pi_{0.5}$ true- and swapped-instruction endpoints as references for the two balls and keep the 314 first frames where they are at least 8 cm apart: DM0.5 heads for the

named ball in 93% of them and, with the swapped instruction, still for the originally named ball in 10%; HY-Embodied does so in 58 and 57%, and LingBot-VLA in 74 and 71% (an earlier LingBot-VLA checkpoint: 55 and 55%). A logistic classifier on the frame-0 layout reaches 53% cross-validated accuracy, near the 52% majority baseline. Together with the weak response to instruction swaps, these results suggest limited instruction conditioning in HY-Embodied and LingBot-VLA; scene-dependent memorization is one possible explanation. The serving and offline paths produce bitwise-identical actions, ruling out discrepancies between these two execution paths.

5.3.3. Where Do Episodes Fail?

Averaged over checkpoints, 21.4% of Core-50 episodes succeed, 13.9% end early, and 64.8% time out, mostly as stalls (Figures 7b and 8): 65–83% of instrumented time-outs record no new checker state after 1× p75. The hands keep moving in a stall: in the last third of time-outs the commanded mean finger angle still reverses 3–4 times per 10 s while its range shrinks from 0.35–0.39 to 0.20–0.28 rad. Acquisition is not the main loss on Core-50. From native checker state we locate most remaining losses at three transitions of multi-finger bimanual control, pooling 800 episodes per task over the eight checkpoints (Figure 5b–d).

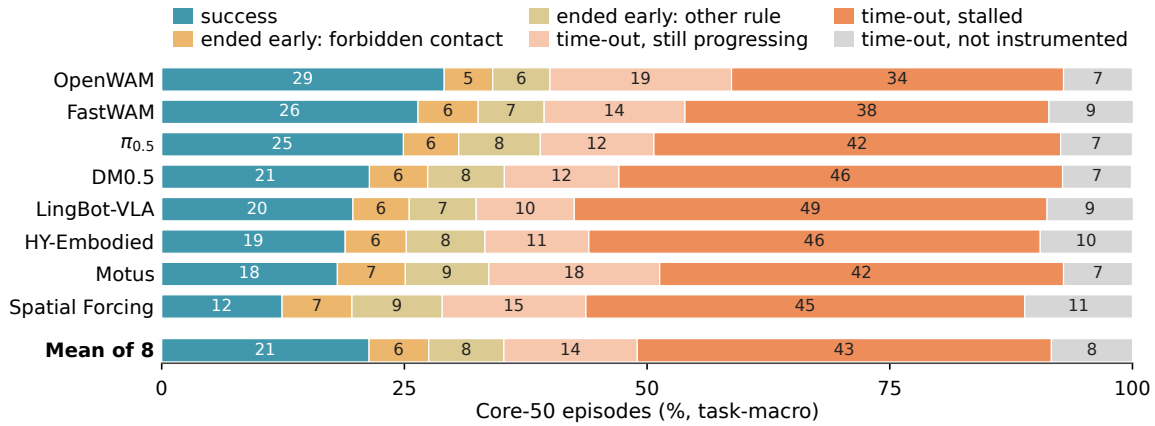


Figure 8 | How Core-50 episodes end, per checkpoint (task-macro share of episodes over the 50 tasks). Early endings are native terminations before the time limit, either forbidden or wrong contact or another rule such as a premature or out-of-order event or a drop. A time-out is stalled when its checker records no new state after the expert 75th-percentile duration and still progressing otherwise; it is not instrumented when the task checker records fewer than two progress states.

Forming a grasp can fire the function. The fingers that hold a tool often rest on its button, trigger, or handles, so closing the grasp can itself trigger an irreversible event. Move and Click Mouse ends as soon as a button is pressed before the mouse has moved 8 cm onto its pad. This ends 443 of 800 episodes, 404 of them before the mouse was ever ready and 297 before it had moved at all (Figure 5b); of the 362 episodes that reach the ready state without an early click, 196 succeed; the checkpoints differ mainly in the early clicks (16 for OpenWAM, 78 for Spatial Forcing). In Photograph Logo the first shutter press precedes aiming in 774 of 778 episodes, and 163 of the 185 episodes that ever aim with the right hand on the shutter succeed. In Water Plant the trigger is a separate finger joint and the grasp is clean, so the loss moves to that finger: of 514 episodes that aim the held sprayer, 238 never spray, 178 of them after moving the trigger at least halfway. In Squeeze Scissors the same fingers must hold and cut: failing episodes close the blades 1,368 times but score 182 cuts, because at 57% of the closings the scissors are not lifted and at 26% a handle contact has been lost. Where

the functional part lies under the holding contacts, grasping fires it early; where it is a separate finger, that finger is not driven to threshold at the right moment; where holding and actuating share fingers, the policy actuates only when it stops holding.

Each side reaches its target, but not at the same time. Set Dual Knobs requires both knobs within 5° of their targets at once, nearly still, held, and then released. Of 800 episodes, the left and right knob reach their target in 411 and 416, both at some point in 219, both at once in 38, and 5 succeed (Figure 5c). When the first knob arrives, the other is still a median 64° away; the first knob then leaves its 7° exit band in 570 of 608 episodes after a median 0.33 s, in 505 cases by overshooting and in 566 with the hand still in contact. The hands turn each knob through its target instead of holding it there while the other hand finishes. Timing alone is not the obstacle: in Pour Seasonings the two streams overlap in 306 of the 398 episodes where both hit their targets.

Reaching a goal and keeping it are different skills. In Open Laptop, Spatial Forcing and OpenWAM bring the display to the requested angle about equally often (73 and 71 of 100 episodes), but it is still there at the end in 23 and 67 (Figure 5d). Of the 154 pooled episodes that lose the angle, 148 end over-opened (median 31.5° past), all with the hand on the display: it is pushed through the target, not dropped. The Cover Pitcher lid stays inserted in 47 of the 86 episodes that insert it, and Stack Blocks keeps four layers in 30 of 164, with the hand already released at 70 of the 134 losses. In Pass Block, the receiver takes over alone in 188 of 367 episodes where both hands hold the block; without takeover the block drops in 175 of 179, a median 0.30 s after the giving hand lets go. Policies must stop pushing at the target and release only when the other hand holds.

Tools are bypassed, and failures are not shared across checkpoints. Line Up Blocks, Scoop Marbles, and Flip Sponge end on direct hand contact with the manipulated object in 81, 44, and 28% of episodes (2 successes in 2,400): policies pursue the object with the hand rather than through the tool. On the 41 fixed-goal tasks at least one checkpoint solves 62.4% of scenes, close to the 64.0% expected under independence (median pairwise $\kappa = 0.03$): the failed scenes differ by checkpoint, so no shared set of hard scenes explains the failures, although each checkpoint may have its own.

Transfer-16 fails at the same transitions, and new objects fail earlier. Every matched transition fails more often than in its Core-50 source (Figure 7c). In Spray Shoe grasps form in 232 of 344 episodes, but at most 3 of 43 per checkpoint complete a trigger cycle. On a solved source, a new object moves failure to acquisition: the left hand picks up the glue stick in 5 of 400 Pass Glue Stick episodes, against 40–88% per checkpoint in Pass Block.

Compositions add a fourth failure: steps are repeated or started early. After the laptop step (146 episodes), no episode brings the mouse onto its pad, and 111 push the display out of range again. The shutter is pressed before the fruit is in the bucket in 311 of 399 Place Fruit and Photograph Logo episodes, water runs in 107 of 400 Scrub Bottle Then Fill Cup episodes although none finishes scrubbing, and the box lid moves before the pitcher lid in 340 of 400 Cover Pitcher and Box episodes. Nothing marks a finished step, so completion must be inferred from the scene.

5.3.4. Task Grouping and Reliability

Table 8 lists the grouping of Core-50 tasks by the hand skill that decides success, used in Figure 5a and throughout Section 5. Each task is assigned to exactly one group by its success condition; tasks

that also require reaching and grasping are assigned to the later skill. The grouping is an analysis lens and is independent of the interaction and structure labels of the catalog. Figure 9 shows the success of every checkpoint on every task in this grouping.

Table 8 | Hand-skill groups of Core-50 tasks. Pooled: successes over all episodes of the eight multi-task checkpoints. Best: mean over tasks of the best checkpoint. Blue: the groups that contain every task above 40% mean success; orange: essentially unsolved groups, where no checkpoint exceeds 21% on any task.

Group (tasks)	Tasks	Pooled %	Best %
Nonprehensile (2)	Rotate Propeller, Roll Ball Through Gate	51.3	62.5
Articulated part (6)	Open Latched Door, Open Laptop, Open Micropipette Box, Open Drawers, Unfold Ruler, Load and Start Microwave	42.9	61.0
Grasp–transport–release (13)	Place Cup on Plate, Place Ball in Bowl, Place Fruit in Bucket, Place Largest Fitting Ball, Place Tableware, Place Wine Glass, Sort Objects, Load Cans, Pass Block, Pass Screwdriver, Store Charger, Cover Box, Lift Carton	34.9	58.0
Finger actuation (9)	Squeeze Scissors, Water Plant, Trigger Drill on Block, Move and Click Mouse, Press Soap Pump, Set Timer, Set Dual Knobs, Photograph Logo, Slide Knife Blade	14.0	26.2
Pour or fill (3)	Pour Seasonings, Fill Cup, Prepare Coffee	13.4	33.0
In-hand reorientation (5)	Orient Cup Upright, Reposition Marker Grasp, Pass and Orient Pen, Sort Nested Cups, Unfold Fan	11.4	20.2
Tool on object (6)	Hammer Nail, Scoop Marbles, Line Up Blocks, Flip Sponge, Scrub Bottle, Stir Pot	2.9	7.5
Precision fit (6)	Insert Peg, Fit Device Cover, Align Blocks in Midair, Connect Charger and Cable, Stack Blocks, Cover Pitcher	2.2	7.3

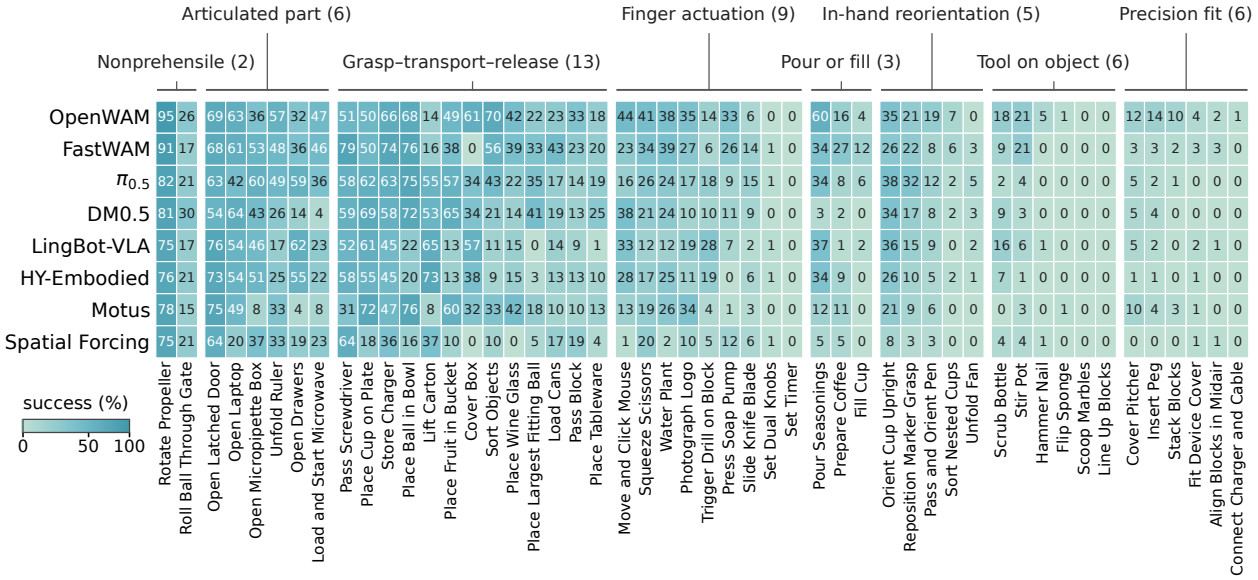


Figure 9 | Core-50 success (%) of the eight multi-task checkpoints on each task, grouped by hand skill as in Table 8 and sorted within a group by mean success. Each cell has 100 episodes (99 for HY-Embodied on Place Wine Glass); DP and task IDs are in Table 14.

Per-task signal-to-noise, the spread of success across checkpoints divided by its binomial standard error, is below 1 on 6 tasks and at least 3 on 14. Twelve tasks selected by this ratio on even-indexed episodes reproduce 96% of the 28 pairwise orders measured on odd-indexed episodes (88% for

random twelve-task subsets), and 24 tasks reproduce all of them. Episode-only 95% half-widths of the 50-task macro score are about 1.9, 1.3, and 0.9 points for 25, 50, and 100 episodes per task. Selecting a checkpoint per task on half of the episodes and scoring it on the other half gives 32.9% versus 29.1% for the best single checkpoint.

An earlier evaluation campaign of the same $\pi_{0.5}$ and OpenWAM checkpoints used the fixed canonical prompt and a single goal per task; on the 48 shared tasks it gives 24.8 versus 23.4% ($\pi_{0.5}$) and 30.5 versus 28.5% (OpenWAM), with per-task Spearman ρ of 0.98 and 0.97 and per-task differences whose standard deviation (4.3 and 6.3 points) is close to the binomial expectation (5.1 and 5.5). Within the Core-50 snapshot, success is not measurably affected by which of the four table materials, three colour palettes, or two roughness levels is drawn (largest within-task effect 0.8 points, permutation $p \geq 0.17$), and training paraphrases score within 2.2 points of the canonical sentence for seven checkpoints (OpenWAM: canonical +4.0 points, $z = 2.9$). All appearance factors and paraphrases are inside the training distribution, so these are invariance checks, not generalization results.

6. Discussion and Conclusion

DexOmni provides a common experimental basis for studying the scope and limits of manipulation competence. By treating embodiment, environment, and task structure as configurable experimental variables, it enables systematic investigation of how policy behavior changes across conditions and where generalization breaks down.

Our evaluation spans more than 40,000 Core-50 episodes and includes eight multi-task checkpoints, revealing substantial differences in performance across task categories. Pooled across checkpoints, success rates reach 35–51% for pushing, articulation, and pick-and-place, but remain at 2–3% for tool use and precision fitting. These disparities persist after accounting for demonstration count and expert demonstration duration. All tasks in both suites have been successfully demonstrated by human operators, making even the hardest cases concrete, achievable targets for policy improvement.

The diagnostic results point to a common requirement: policies must regulate the state of an interaction throughout execution. They need to distinguish contacts that stabilize an object from those that actuate it, condition release on whether the partner's hand has secured the object, and maintain achieved goals as the task progresses. Failures on composed tasks further highlight the importance of tracking completed steps and satisfying prerequisites before advancing. Together, these findings motivate learning methods that explicitly account for contact roles, inter-hand dependencies, and task progress.

Future evaluations should measure both task completion and whether policies maintain the conditions needed for success throughout execution. DexOmni provides a foundation for testing whether improvements in these aspects of control generalize across embodiments and translate into dependable behavior on physical robots. Future work could expand embodiment coverage and longer-horizon compositional tasks, strengthen paired simulation and real-world evaluation, and investigate policies that jointly model contact state, inter-hand coordination, and task progress. These directions would help establish which advances produce transferable manipulation capabilities and under what conditions they remain reliable.

7. Future Work

Tactile sensing. Every DexOmni task is solved from vision and proprioception alone, but the wrist cameras are mounted on the hand and the held object occludes them: once a grasp forms, the very region the policy must control leaves the view. The failures we report sit exactly there. A grasp that closes over a button hides the button as it presses it; a cut that loses a handle contact happens on the

far side of the object; a goal pushed past its target is obscured by the hand doing the pushing. Tactile sensing would observe these contact events directly instead of inferring them from a partially hidden view. Adding it requires taxels placed identically on the simulated and physical hand, a contact model verified rather than assumed, and tasks that still admit a visual solution, so that visual and tactile policies remain comparable on the same predicates.

Quantified physical evaluation. The real-robot rollouts in Appendix E show that simulation-trained checkpoints act sensibly on the physical platform, but they are recorded evidence, not a measurement. A physical success rate needs the discipline the simulated suite already has: paired simulation and real instances, an executable predicate on both, a fixed trial count, and reported failure modes rather than selected successes. The protocol is specified in Appendix B.5; what remains is the trial budget to run it alongside simulation-plus-real and real-only training under an explicit data budget.

Long-horizon household and production tasks. DexOmni scores individual skills, and even the compositional Transfer-16 tasks chain at most a few operations. Household and production work is longer and less forgiving: tens of operations over a changing world state, objects found rather than placed, and success measured in throughput and rework rather than a binary predicate. Our failure analysis indicates what such a suite must measure beyond success, namely whether a policy tracks which steps are complete, notices that an earlier step went wrong, and can recover without restarting the episode. That requires long-horizon demonstrations, a stage model that admits recovery rather than enforcing an ordered prefix, and evaluation that reports progress and interventions.

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A. Task Inventory and Taxonomy

Table 9 lists all 66 tasks in DexOmni: Core-50 provides 50 tasks for learning and evaluation; Transfer-16 provides 16 additional tasks exclusively for generalization evaluation. Transfer-16 contains 6 compositional tasks (EVAL-Comb), 7 generalization tasks (EVAL-Gen), and 3 tasks in the EVAL-Flex family. Evaluation-only tasks are excluded from the dataset for policy training. Within each split, task identifiers follow alphabetical order of the English task names.

Source superscripts distinguish task-design adaptation (J: DexJoCo; V: DexVerse) from asset reuse (JA and VA, respectively). The sources and modifications are detailed in Appendix A.1. Unmarked entries do not imply that every asset or component is original.

The three label columns are the axes of the taxonomy in Figure 3. Interaction Type records the dominant manipulation requirement. Asset Type includes all applicable types of task-relevant assets, including designated targets and receptacles, according to their current simulation models and active degrees of freedom. The robot, generic support surfaces, and background assets are excluded. Task Structure describes the organization of semantic goals and operations; approaching, grasping, and releasing do not by themselves create separate stages.

Table 9 | Complete task inventory and three-axis taxonomy of DexOmni; shaded rows are the evaluation-only Transfer-16 tasks. Abbreviations are defined below the table.

Task ID	Task Name	Interaction Type	Task Structure	Asset Type
01	Align Blocks in Midair	CR	SS	RB
02	Connect Charger and Cable	CR	LH	RB
03	Cover Box	CR	SS	RB
04	Cover Pitcher	CR	SS	RB
05	Fill Cup	F	LH	RB, AR
06	Fit Device Cover	CR	SS	RB
07	Flip Sponge	F	SS	RB
08^{J}	Hammer Nail	F	SS	RB, AP
09^{JA}	Insert Peg	CR	SS	RB
10^{V}	Lift Carton	CR	SS	RB
11	Line Up Blocks	F	RP	RB
12^{J}	Load and Start Microwave	F	LH	RB, AR, AP
13VA	Load Cans	P	SS	RB
14^{J}	Move and Click Mouse	F	LH	RB, AP
15	Open Drawers	P	SS	AP
16^{V}	Open Laptop	P	SS	AR
17	Open Latched Door	F	LH	AR
18	Open Micropipette Box	P	SS	AR
19	Orient Cup Upright	P	SS	RB
20	Pass and Orient Pen	CR	LH	RB
21	Pass Block	P	LH	RB
22	Pass Screwdriver	P	LH	RB
23^{J}	Photograph Logo	F	SS	RB, AP
24	Place Ball in Bowl	P	SS	RB
25	Place Cup on Plate	P	SS	RB

Continued on next page

Table 9 | Complete task inventory and three-axis taxonomy of DexOmni (continued).

Task ID	Task Name	Interaction Type	Task Structure	Asset Type
26	Place Fruit in Bucket	P	SS	RB
27	Place Largest Fitting Ball	P	SS	RB
28	Place Tableware	P	SS	RB
29	Place Wine Glass	P	SS	RB
30	Pour Seasonings	F	SS	RB
31	Prepare Coffee	F	LH	RB, AR, AP
32	Press Soap Pump	F	SS	RB, AP
33	Reposition Marker Grasp	CR	SS	RB
34	Roll Ball Through Gate	P	SS	RB
35	Rotate Propeller	CR	RP	AR
36	Scoop Marbles	F	SS	RB
37	Scrub Bottle	CR	RP	RB
38	Set Dual Knobs	F	SS	AR
39	Set Timer	F	RP	AP
40^{V}	Slide Knife Blade	CR	SS	AP
41	Sort Nested Cups	CR	LH	RB
42	Sort Objects	P	LH	RB
43	Squeeze Scissors	F	RP	AR
44	Stack Blocks	P	RP	RB
45	Stir Pot	F	RP	RB
46VA	Store Charger	P	LH	RB, AP
47	Trigger Drill on Block	F	SS	RB, AP
48	Unfold Fan	CR	SS	AR
49	Unfold Ruler	CR	SS	AR
50^{J}	Water Plant	F	SS	RB, AR
EVAL-Comb01	Cover Pitcher and Box	CR	RP	RB
EVAL-Comb02	Open Laptop and Click Mouse	F	LH	RB, AR, AP
EVAL-Comb03	Pass Block and Close Box	CR	LH	RB
EVAL-Comb04	Place Fruit and Photograph Logo	F	LH	RB, AP
EVAL-Comb05	Scrub Bottle Then Fill Cup	F	LH	RB, AR
EVAL-Comb06	Season and Stir	F	LH	RB
EVAL-Gen01	Cover Lunchbox	CR	SS	RB
EVAL-Gen02	Open Trash Bin	P	SS	AR
EVAL-Gen03	Orient Bottle Upright	P	SS	RB
EVAL-Gen04	Pass Glue Stick	P	LH	RB
EVAL-Gen05	Set Mixer Knobs	F	RP	AR
EVAL-Gen06	Spray Shoe	F	SS	RB, AR
EVAL-Gen07	Stack Boxes	P	RP	RB
EVAL-Flex01	Compress Sponge into Mug	CR	SS	RB, DF
EVAL-Flex02	Fold Towel Twice	CR	RP	DF
EVAL-Flex03	Pack Pillow	CR	SS	RB, DF

Abbreviations. **Interaction Type:** P = Primitive; F = Functional; CR = Contact-rich. **Task Structure:** SS = Single-stage; RP = Repetitive; LH = Long-horizon (multi-stage). **Asset Type:** RB = Rigid body; AP = Articulated (prismatic); AR = Articulated (revolute); DF = Deformable.

Classification notes. Primitive covers object transport or reorientation whose success is decided by reaching a target configuration, including pushing, pulling, or rolling an object and moving an articulated part along its free degree of freedom (e.g., opening a drawer or a hinged lid). Functional requires the purposeful use of a tool, appliance, or control interface (buttons, knobs, levers, faucets, triggers) to produce an effect. Contact-rich denotes tasks dominated by fine contact coordination, such as regrasping, precise mating, sustained contact-force regulation, coordinated bimanual articulation, or controlled deformation. RB describes each non-articulated rigid asset and does not limit a task to one object. Rigid links of articulated assets are not counted separately. Deformable labels follow the current simulation model; visual liquid effects do not add a DF label. In Task 08, the nail has a prismatic joint relative to the wood block, and the hammer is a separate rigid object. EVAL-Flex03 combines a deformable foam pillow with a rigid carton. SS denotes a single semantic goal. RP denotes repeated applications of the same kind of manipulation operation, whereas LH requires distinct, dependent subgoals. Routine setup and terminal activation, release, or waiting do not by themselves turn a repetitive task into LH. Task 39 (Set Timer) is classified as RP because its main manipulation consists of setting the minute and second values through button operations; starting the timer, withdrawing the hands, and observing a countdown tick are terminal completion conditions.

A.1. Task Provenance and Adaptation

DexOmni incorporates selected task designs and assets from DexJoCo [1] and DexVerse [2]. Table 10 separates task-design lineage from the reuse of particular assets. The J and V markers identify documented adaptations; JA and VA identify asset provenance without asserting that the whole task was transferred. Shared infrastructure and generic similarities, such as grasping or stacking, are insufficient to establish task-specific lineage.

Compositions and Transfer-16. Source credit also follows reused components into new evaluations. EVAL-Comb02 combines the adapted laptop and mouse skills, EVAL-Comb04 reuses the photograph skill, and EVAL-Gen06 transfers the spray interaction from Water Plant. These are compositions or task variants of the credited components, rather than entire tasks taken from an upstream benchmark. The evaluation-only designation specifies exclusion from policy-training demonstrations; it does not imply that all constituent skills or assets are new.

Attribution boundaries. Task names alone do not establish reuse. For example, Prepare Coffee and Squeeze Scissors have related counterparts in DexVerse but different documented workflows or asset lineages; we do not label them as direct ports on that basis. The source markers identify documented task-design adaptations and retained asset reuse.

Table 10 | Acknowledged task-design and asset sources. IDs follow Table 9. All listed tasks use our platform's execution and observation pipeline; source attribution does not imply identical reset distributions or success predicates.

ID	Upstream task or asset	Adaptation or reuse in DexOmni
08 J	DexJoCo: Hammer nail	Adds a supporting hand holding the wood block and a task-specific nail-depth condition.
12 J	DexJoCo: Bimanual microwave	Adapts the loading, door-closing, and activation sequence to the collection platform.
14 J	DexJoCo: Click mouse	Requires moving the mouse onto its pad before a permitted click.
23 J	DexJoCo: Bimanual photograph	Retains camera aiming and shutter activation with platform-specific contact and execution logic.
50 J	DexJoCo: Water plant	Adds grasp, trigger-contact, and aim-sensitive spray conditions.
10 V	DexVerse: BimanualLiftCarton	Adapts bilateral lifting to lower-side support and a three-second airborne hold.
16 V	DexVerse: OpenLaptop	Adapts the opening task to the ROKAE–Wuji platform; the released task uses a replacement laptop asset and a 60-degree target.
40 V	DexVerse: SlideUtilityKnife	Adapts blade sliding to coordinated hand roles and target-conditioned motion; the collection asset was replaced.
09 JA	IndustReal peg/socket via DexJoCo	Reuses the peg/socket asset family with a separately implemented single-arm insertion task.
13 VA	YCB soup can via DexVerse	Reuses the can asset in a new synchronized cross-loading task with a project-authored partitioned bin.
46 VA	Drawer and charger via DexVerse	Reuses the drawer mechanism and charger; adds a replacement cabinet appearance and a hold–store–release sequence.

B. Additional Data, Platform, and Protocol Records

B.1. Physical System and Alignment Record

Figure 2a compares simulated and physical views of the toy-aircraft setup from the setup, wrist, and main cameras.

The ROKAE AR5-08 has seven degrees of freedom, a rated payload of 5 kg, a 683 mm reach, and manufacturer-specified repeatability of ±0.1 mm [29]. The Wuji Hand has 20 actively actuated joints, four per finger, with position control over USB 2.0 [30]. Its specified 1 kHz low-level frequency is distinct from policy inference and host command rates.

The correspondence record covers hand/robot geometry and limits, asset scale, camera calibration, state/action coordinate frames, joint/side ordering, timing, initial conditions, and success. Calibration data are disclosed independently of target-task demonstrations.

B.2. Transfer-16 Duration Map

Table 11 reports the 30 Hz protocol. The source-frame column sums each mapped atomic task's P75 before multiplying by three. Task validity is established by the verification in Section 3.2; a repaired task requires fresh trials under its qualified implementation. Table 9 lists the paper task IDs and names.

Table 11 | Transfer-16 physical-time budgets. Flex budgets (shaded) are native exceptions and are not described as P75 estimates.

Task ID	Supporting Core-50 tasks	P75 frames	Seconds	Actions
Comb01	Cover Pitcher + Cover Box	382 + 210	59.2	1776
Comb02	Open Laptop + Move and Click Mouse	196 + 125	32.1	963
Comb03	Pass Block + Cover Box	195 + 210	40.5	1215
Comb04	Place Fruit in Bucket + Photograph Logo	110 + 158	26.8	804
Comb05	Scrub Bottle + Fill Cup	259 + 530	78.9	2367
Comb06	Pour Seasonings + Stir Pot	236 + 547	78.3	2349
Gen01	Cover Box	210	21.0	630
Gen02	Open Micropipette Box	125	12.5	375
Gen03	Orient Cup Upright	74	7.4	222
Gen04	Pass Block	195	19.5	585
Gen05	Set Dual Knobs	431	43.1	1293
Gen06	Water Plant	139	13.9	417
Gen07	Stack Blocks	387	38.7	1161
Flex01	Native exception: Compress Sponge into Mug	–	240	7200
Flex02	Native exception: Fold Towel Twice	–	180	5400
Flex03	Native exception: Pack Pillow	–	180	5400

B.3. Task Partitions and Generalization Design

A task is defined by its environment, initial-state distribution, observations, actions, success predicate, and physical time budget:

$$\mathcal{T}_i = (\mathcal{E}_i, p_i(s_0), \mathcal{O}_i, \mathcal{A}_i, G_i, H_i). \tag{2}$$

We distinguish training trajectories, held-out trajectories from training tasks, fresh closed-loop trials on those tasks, and evaluation-only tasks. Trajectory validation measures action prediction; fresh closed-loop trials measure execution under the task's native training-source distribution. Transfer-16 tasks contribute no demonstrations to benchmark training or task-specific finetuning. Their absence from this dataset does not establish their absence from a foundation model's external pretraining.

The evaluation catalog contains six EVAL-Comb, seven EVAL-Gen, and three EVAL-Flex tasks. These implementation groups support complementary generalization questions rather than a set of perfectly orthogonal factors. Each task must specify the supporting training skills and assets, the held-out change, and its executable goal. Multiple changes are identified when they co-occur.

Asset and relation transfer. This dimension tests the use of skills with new objects, asset pairings, or goal relations. An unseen combination of familiar assets is distinguished from a new instance or category. For example, covering a new container and orienting a different object test transfer of related functional behavior, but whether an asset is actually familiar is determined by source and asset identities, not by its task name.

Skill transfer and composition. These tasks reuse skills in new sequences or contexts. *Open Laptop and Click Mouse* combines opening an articulated object with interface operation; *Scrub Bottle Then Fill Cup* connects tool interaction to a subsequent goal. Evaluation records where an execution stops and whether a successful stage leaves the state needed by the next stage. A previously unseen atomic operation is identified separately from a new composition of trained skills.

Table 12 | Training support and evaluation changes. Relations shown here describe task semantics; the source/asset mapping specifies actual exposure.

Dimension	Training support	Evaluation question
Asset/relation	Object interaction, covering, orienting, passing	Can a skill operate on a new target or asset relation?
Skill composition	Opening a laptop; moving and clicking a mouse; scrubbing; filling	Can trained behaviors be connected in a new sequence with valid stage transitions?
Deformable tasks	Rigid/articulated object manipulation	Can the policy adapt to compression, folding, and packing?

Transfer to deformable manipulation. *Compress Sponge into Mug*, *Fold Towel Twice*, and *Pack Pillow* require handling deformation without deformable-task demonstrations in the benchmark training set. These are transfer challenges across object, goal, and physical behavior. Isolating compliance as a causal factor would additionally require matched geometry and goals; the present suite does not assume that all differences are due to material properties.

B.4. Visual and Language Robustness Protocols

Training randomization and evaluation under domain shift serve different roles. Fresh samples from the training distribution measure in-distribution performance; held-out combinations or expanded ranges of visual factors measure robustness under specified shifts. These tests retain the task goal and identify the factors changed. They are reported separately from the evaluation-only tasks, which change asset relations, skills, or physical interactions. Combining both changes is an additional challenge condition, not a replacement for the matched-distribution baseline.

The eight held-out paraphrases per active goal define a language-only test: the environment, target values, reset seeds, and success predicate stay fixed while the instruction changes. Paired trials compare canonical, training-paraphrase, and held-out-paraphrase success. We report the success gap and paired outcome agreement, so consistently failing all wordings cannot masquerade as successful language robustness. This test is separate from Transfer-16 physical tasks and from visual domain shifts.

A complementary goal-sensitivity test changes the requested object, relation, or numerical setting only within a task's supported configurations, with its executable goal updated accordingly. For example, the timer must match the requested minutes and seconds; moving a knob must reach the requested angle and direction. Such tests assess whether the policy uses the instruction, rather than repeating a familiar motion. Unseen numerical values require an explicitly held-out goal split. Text-invariant policies are not scored for language robustness; their structured-goal performance is reported separately. These additional tests require dedicated trials and are not inferred from the canonical-prompt evaluations.

B.5. Matched Physical-Transfer Protocol

A paired task comprises simulation and real instances with a correspondence record:

$$\mathcal{P}_i = (\mathcal{T}_i^{\text{sim}}, \mathcal{T}_i^{\text{real}}, C_i). \tag{3}$$

The record specifies robot and camera configuration, state/action frames, control timing, initialization conditions, and success definitions. Direct transfer deploys the same checkpoint without target-task real demonstration training or real-data finetuning. Calibration measurements and any real images used to configure the system are disclosed separately from policy training. External pretraining of a VLA or WAM retains its own data identity.

Only tasks with physical counterparts enter the paired analysis, which reports their coverage and per-task simulation and real trial counts. The signed success-rate difference for model m and task i is defined as

$$\Delta_{i,m} = S^{\text{sim}}_{i,m} - S^{\text{real}}_{i,m}. \tag{4}$$

Same-task real transfer and Transfer-16 generalization are independent claims. The recorded real-robot rollouts of this platform are reported qualitatively (Appendix E) and do not constitute a scored physical evaluation.

B.6. Additional Task-Capability Examples

Articulated collection assets occur in 21 tasks: 12 involve revolute joints and 11 prismatic joints, with two containing both. Goals specify motion direction, travel, angle, or unlocking prerequisites. Set Timer uses requested minutes and seconds; Set Dual Knobs uses left/right angle and direction targets; and Unfold Fan targets 60, 90, or 120 degrees. Fixed three-cycle scissors and propeller requirements do not by themselves establish unseen-count generalization. Place Tableware combines identity, position, and orientation; Place Largest Fitting Ball and Sort Nested Cups add size comparisons. Prepare Coffee and Load and Start Microwave require intermediate states that enable later operations. These capability labels overlap the catalog's 17 primitive, 18 functional, and 15 contact-rich tasks and its 31 single-stage, seven repetitive, and 12 long-horizon tasks.

C. Core-50 Results and Coverage

Table 5 is computed from 45,000 episodes on all 50 Core-50 tasks. The eight multi-task checkpoints contribute 100 episodes per task (39,999 valid; HY-Embodied has 99 valid episodes on Place Wine Glass), and single-task DP contributes 5,000 episodes from task-specific checkpoints. Overall success is the unweighted mean of the 50 task success rates; interaction-type and hand-skill means use the same rule within each group. Chinese Chess remains outside the catalog.

Scope and validity. All multi-task evaluations passed the recorded runtime validation and match the selected checkpoints; the DP checks are described below. The evaluator samples each reset from the admitted source templates of its task, so the scores describe this implemented reset distribution. Native reset rejection is handled before policy scoring by selecting the first accepted candidate while retaining the assigned goal and language; rejected resets and infrastructure interruptions are not policy failures. On 41 tasks, the eight multi-task checkpoints share the reset seed, variant, goal, appearance, and prompt at nearly every episode index; on the nine goal-conditioned tasks, the goal and prompt are sampled per checkpoint.

Replacement and scoring records. Exact original-slot associations include the recovered minority-goal Fit Device Cover episodes and the recovered DM0.5 Place Wine Glass episode; no slot is counted twice. Each checkpoint contributes 100 Lift Carton episodes evaluated using the following criteria: both hands support opposite lower sides of the carton, the entire carton clears the table, and it remains continuously airborne for three seconds.

Single-task DP reference. DP uses a distinct checkpoint at 30,000 updates for each task. Ten tasks use completed retraining with corrected near-degenerate normalization; the other tasks retain their previously selected task checkpoints. Lift Carton is evaluated in a separate batch using the lift/hold criteria described above; its result is included in the 50-task score but excluded from the

11-task matched comparison. Original and retrained model identities are preserved per task. The 38 earlier evaluations use the earlier native goal sampler, whereas the newer runs use frozen training-frame-weighted goal quotas, so DP is a descriptive reference rather than a goal-distribution-matched comparison. Three rejected knife-task resets are omitted under the original first-100-accepted-seeds rule. The DP audit checks checkpoint hashes, native completion, native success, and recorded three-view video completion.

The accompanying machine-readable record stores the task map, checkpoint identities, source plans, supplemental-slot associations, and the hash of the episode-level snapshot.

Uncertainty. Table 13 reports two 95% percentile bootstrap intervals. The episode-only interval (20,000 resamples) redraws each task's outcomes from its observed success rate with the 50 tasks fixed; it quantifies finite rollout sampling under an independent-episode assumption. The task-and-episode interval (2,000 resamples) also resamples the 50 tasks with replacement, so it describes a new draw of tasks from the same catalog. Neither interval includes training-run or checkpoint-selection uncertainty.

Table 13 | Core-50 task-macro success on all 50 tasks with 95% bootstrap intervals over episodes only and over tasks and episodes. Shaded: best.

Model	Successes / episodes	SR (%)	Episodes CI (%)	Tasks+episodes CI (%)
OpenWAM	1,453 / 5,000	**29.1**	[28.0, 30.1]	[22.8, 36.0]
FastWAM	1,319 / 5,000	26.4	[25.4, 27.4]	[20.0, 33.8]
$\pi_{0.5}$	1,243 / 5,000	24.9	[23.8, 25.9]	[18.6, 31.7]
DM0.5	1,072 / 5,000	21.4	[20.5, 22.4]	[15.2, 28.1]
LingBot-VLA	984 / 5,000	19.7	[18.8, 20.6]	[13.6, 26.2]
HY-Embodied	947 / 4,999	18.9	[18.0, 19.8]	[13.2, 25.4]
Motus	903 / 5,000	18.1	[17.2, 18.9]	[12.3, 25.0]
Spatial Forcing	620 / 5,000	12.4	[11.6, 13.2]	[7.9, 18.0]
DP (single-task)	483 / 5,000	9.7	[9.0, 10.3]	[5.0, 14.9]

Table 14 | Per-task success rates (%) on the 50 Core-50 tasks. Each cell has 100 episodes, so it also gives the number of successes; *HY-Embodied has 99 valid episodes on Place Wine Glass. Columns use the selected checkpoints from Section 5.1; OW, FW, LB, HY, and SF denote OpenWAM, FastWAM, LingBot-VLA, HY-Embodied, and Spatial Forcing. The DP sampler caveat above applies to every DP cell. Shaded: best on each task, ties included.

ID	Task	OW	FW	$\pi_{0.5}$	DM0.5	LB	HY	Motus	SF	DP
01	Align Blocks in Midair	2	3	0	0	1	0	0	1	0
02	Connect Charger and Cable	1	0	0	0	0	0	0	0	0
03	Cover Box	61	0	34	34	57	38	32	0	0
04	Cover Pitcher	12	3	5	5	5	1	10	0	2
05	Fill Cup	4	12	6	0	2	0	0	0	0
06	Fit Device Cover	4	3	0	0	2	1	1	1	0
07	Flip Sponge	1	0	0	0	0	0	1	0	0
08	Hammer Nail	5	0	0	0	1	0	0	1	0
09	Insert Peg	14	3	2	4	2	1	4	0	0
10	Lift Carton	14	16	55	53	65	73	8	37	51
11	Line Up Blocks	0	0	0	0	0	0	0	0	0
12	Load and Start Microwave	47	46	36	4	23	22	8	23	0
13	Load Cans	23	43	17	19	14	13	10	17	0
14	Move and Click Mouse	44	23	16	38	33	28	13	1	29
15	Open Drawers	32	36	59	14	62	55	4	19	29
16	Open Laptop	63	61	42	64	54	54	49	20	23
17	Open Latched Door	69	68	63	54	76	73	75	64	53
18	Open Micropipette Box	36	53	60	43	46	51	8	37	46
19	Orient Cup Upright	35	26	38	34	36	26	21	8	38
20	Pass and Orient Pen	19	8	12	8	9	5	6	3	0
21	Pass Block	33	23	14	13	9	13	10	19	2
22	Pass Screwdriver	51	79	58	59	52	58	31	64	23
23	Photograph Logo	35	27	17	10	19	11	34	10	4
24	Place Ball in Bowl	68	76	75	72	22	20	76	16	6
25	Place Cup on Plate	50	50	62	69	61	55	72	18	25
26	Place Fruit in Bucket	49	38	57	65	13	13	60	10	6
27	Place Largest Fitting Ball	22	33	35	41	0	3	18	5	0
28	Place Tableware	18	20	19	25	1	10	13	4	0
29	Place Wine Glass*	42	39	22	14	15	15	42	0	9
30	Pour Seasonings	60	34	34	3	37	34	12	5	1
31	Prepare Coffee	16	27	8	2	1	9	11	5	0
32	Press Soap Pump	33	26	9	11	7	0	1	12	0
33	Reposition Marker Grasp	21	22	32	17	15	10	9	3	5
34	Roll Ball Through Gate	26	17	21	30	17	21	15	21	8
35	Rotate Propeller	95	91	82	81	75	76	78	75	79
36	Scoop Marbles	0	0	0	0	0	0	0	0	0
37	Scrub Bottle	18	9	2	9	16	7	0	4	22
38	Set Dual Knobs	0	1	1	0	1	1	0	1	0
39	Set Timer	0	0	0	0	0	0	0	0	0
40	Slide Knife Blade	6	14	15	9	2	6	3	6	0
41	Sort Nested Cups	7	6	2	2	0	2	0	0	0
42	Sort Objects	70	56	43	21	11	9	33	10	0
43	Squeeze Scissors	41	34	26	21	12	17	19	20	2
44	Stack Blocks	10	2	1	0	0	0	3	0	0

Continued on next page

Table 14 | Per-task success rates (continued).

ID	Task	OW	FW	$\pi_{0.5}$	DM0.5	LB	HY	Motus	SF	DP
45	Stir Pot	21	21	4	3	6	1	3	4	0
46	Store Charger	66	74	63	58	45	45	47	36	9
47	Trigger Drill on Block	14	6	18	10	28	19	4	5	2
48	Unfold Fan	0	3	5	3	2	1	0	0	0
49	Unfold Ruler	57	48	49	26	17	25	33	33	2
50	Water Plant	38	39	24	24	12	25	26	2	7
Task-macro mean		29.1	26.4	24.9	21.4	19.7	18.9	18.1	12.4	9.7

D. Single-Task Demonstration Scaling

Training controls. Tasks 24 and 43 use independent $\pi_{0.5}$ finetuning runs initialized from the same generic pretrained base with a freshly initialized 52-dimensional action interface. All runs use full finetuning, batch size 32, training seed 0, and exactly 20,000 optimizer updates. The learning rate warms up for 1,333 updates to 5×10^{-5} and decays to 5×10^{-6}. The 10/50/100/200-trajectory subsets are a single frozen nested draw; sampling for Task 24 is stratified approximately in proportion to the three ball types. The full tier contains every task-local training trajectory: 438 for Task 24 (192 tennis, 175 mini soccer, 71 golf) and 439 for Task 43. The 62 held-out trajectories for Task 24 and 61 for Task 43 are excluded at every size. Images, augmentation, state/action semantics, the prompt bank, and the 20/80 canonical/paraphrase training sampler are unchanged across sizes.

Task-wide normalization statistics are fixed across sizes and computed from the full training pool, excluding validation. Thus small-data runs have access to larger-pool summary statistics; this is not a strictly subset-only data-access experiment. Fixed update counts also imply different reuse rates across dataset sizes. The selected checkpoint is always update 20,000, not the checkpoint with the highest test success. Full-inference offline validation runs every 1,000 updates on fixed anchors; the held-out trajectories for Task 24 contain golf balls only, so this validation does not represent the complete ball distribution. The curves here report fresh closed-loop success, not offline reconstruction error.

Closed-loop evaluation. Each checkpoint has 100 accepted, runtime-valid episodes, giving 1,000 episodes in total with no missing slots. Sizes share the same planned task-specific reset seeds, goals, and prompts. Native reset rejection may advance an accepted seed; shared planned draws do not certify identical complete physical states. Evaluation keeps the native EEF controller, 30 Hz action execution, the accepted rendering revision, and checkpoint normalization. The physical budgets are 3×P75: 321 actions (10.7 s) for Task 24 and 663 (22.1 s) for Task 43. Inference uses a 30-action horizon and ten denoising steps. The prompt bank and goal distribution are fixed across training sizes rather than refitted to each subset. Task 24 evaluates 42 tennis, 40 mini-soccer, and 18 golf goals; Task 43 requires lifting the scissors and completing three cuts in the air.

Figure 7 and Table 15 use raw success fractions under these fixed evaluation draws. Reweighting the per-goal success of Task 24 by the frozen training-frame prior instead gives 6.99%, 22.89%, 30.88%, 64.67%, 82.80%, in increasing data-size order; Task 43 has one goal and is unchanged. The intervals describe approximate rollout-level binomial uncertainty conditional on each checkpoint; they do not estimate training-seed or subset-selection variance and are not simultaneous confidence bands. There is only one training seed and one subset draw at each size.

Table 15 | Single-task scaling results. Success counts are out of 100 trials per cell; intervals are pointwise 95% Wilson intervals in percent. Full-pool sizes are given for Tasks 24 and 43, respectively. Shaded: best.

Trajectories	Task 24 s/n	95% CI	Task 43 s/n	95% CI
10	7/100	[3.4, 13.7]	1/100	[0.2, 5.4]
50	23/100	[15.8, 32.2]	11/100	[6.3, 18.6]
100	31/100	[22.8, 40.6]	13/100	[7.8, 21.0]
200	65/100	[55.3, 73.6]	18/100	[11.7, 26.7]
Full (438 / 439)	**83/100**	[74.5, 89.1]	**44/100**	[34.7, 53.8]

Result scope and provenance. The runtime checks pass. As for Core-50 (Appendix C), the evaluator reuses admitted source templates, so these curves describe the implemented closed-loop reset distribution rather than every training-source cohort. Machine-readable counts, result hashes, checkpoint steps, goal strata, and evidence-manifest hashes accompany the manuscript in `data/scaling_results_20260926.json`.

E. Qualitative Real-Robot Rollouts

The videos released with this work show real-robot rollouts of checkpoints adapted using this dataset's simulation demonstrations, without target-task real-robot demonstrations (Section 4.5); they are available at https://github.com/lllttwww/DexOmni. Two task families are shown: single-hand reach-and-place interactions and bimanual tasks that require both hands to hold their targets at once.

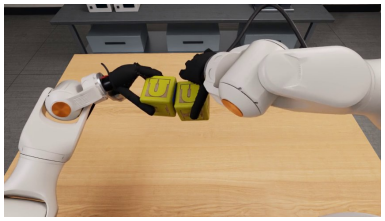
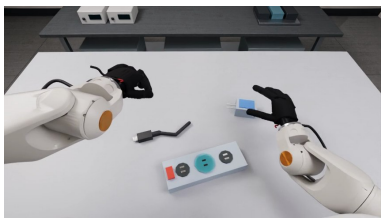
We include these recordings as qualitative evidence that the simulated demonstrations transfer to a physical dexterous hand: the policies establish contact with the intended object, carry it, and preserve the object pose through the hand transitions that the simulated episodes also rely on. The recordings are not a scored evaluation. They cover a small number of episodes on one physical platform, they are not paired with a controlled simulation-plus-real or real-only training condition under a matched data budget, and no success rate is reported. We therefore do not use them to quantify physical transfer, and we list a budget-controlled version of this comparison as future work (Section 7).

F. Task Profiles

Table 16 follows the task order in Table 9. Each row presents the task identifier and name, a screenshot, and the task instruction; the interaction, task-structure, and asset labels of each task are listed in Table 9. Tasks 01–50 are used for both training and evaluation; the 16 tasks with EVAL-prefixed identifiers form Transfer-16. Source superscripts match Table 9; Appendix A.1 explains task adaptations and asset reuse. For training tasks, we show one canonical prompt from the training set; for Transfer-16 tasks, we show the configured task prompt. When a task has multiple target-specific configurations, one instance is shown.

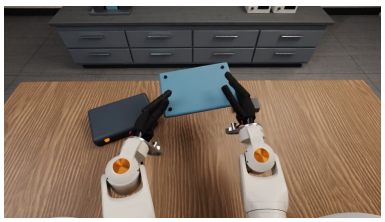
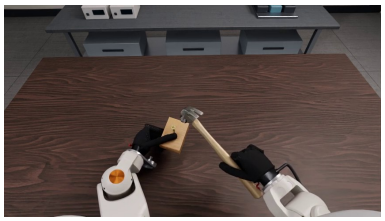
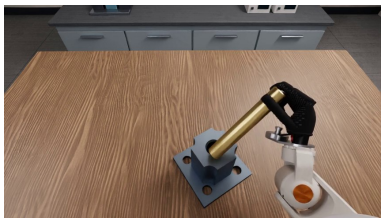
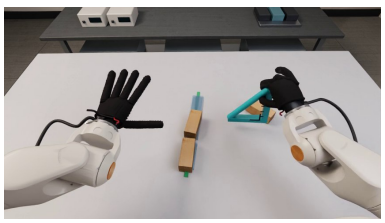
Within the Task Instruction column, **bold text** marks the randomly initialized task parameters, such as target objects, spatial relations, orientations, angles, durations, and action choices. Concrete values show one example configuration. Visual randomization that does not alter the instruction is not marked. Meaning-preserving prompt variants are a separate language dimension (Section 4.4); the instructions here do not enumerate the 32 training and eight held-out paraphrases associated with each goal-specific canonical instruction.

Table 16 | Task profiles of DexOmni.

Task	Screenshot	Task Instruction
01 Align Blocks in Midair		Lift both assigned blocks with their assigned hands, then mate their marked faces in midair and hold the physical contact stable for 0.4 seconds.
02 Connect Charger and Cable		Insert the charger plug into the power strip first, then insert the charging cable connector into the charger head.
03 Cover Box		Align the box lid and place it onto the clear box.
04 Cover Pitcher		Align the pitcher lid and place it onto the pitcher body.

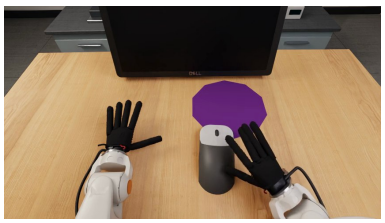
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Table 16 | Task profiles of DexOmni (continued).

Task	Screenshot	Task Instruction
05 Fill Cup		Hold the mug upright under the faucet with the left hand. Open the right lever outward past 70 percent, fill for half a second, place the mug on the marked pad, and close the faucet.
06 Fit Device Cover		Use both hands on opposite sides of the device cover to carry it together over the device body, then align it with its matching interface and seat it.
07 Flip Sponge		Hold the skillet level with the left hand. Use the right-hand spatula to lift and flip the sponge once, leaving its opposite face stable inside the skillet. Do not touch the sponge directly.
08 Hammer Nail[J]		Hold the wood block with the left hand, keeping its nail inserted. Pick up the hammer with the right hand and strike the nail so it sinks into the block.
09 Insert Peg[JA]		Insert the round peg into the socket tray hole.
10 Lift Carton[V]		Support opposite lower carton sides with both hands, lift the entire carton clear of the table, and keep it continuously airborne for 3 seconds.
11 Line Up Blocks		Use only the set square tool to push three scattered wood blocks onto the baseline and align them side by side.

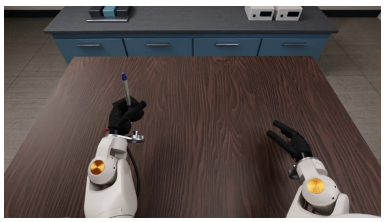
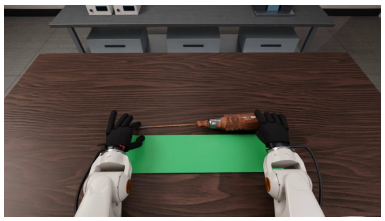
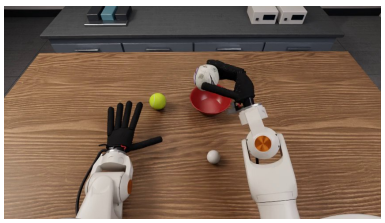
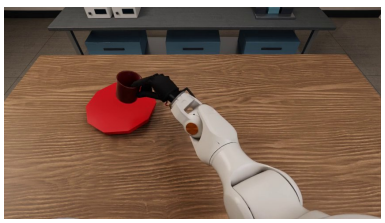
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Table 16 | Task profiles of DexOmni (continued).

Task	Screenshot	Task Instruction
12 Load and Start Microwave[J]		Open the microwave door, carry the plate with its food box into the microwave, close the door, then press the green start button.
13 Load Cans[VA]		Pick up both soup cans with their assigned hands and cross-load them into the opposite bin partitions. Release them within the synchronization window and leave them stable.
14 Move and Click Mouse[J]		Move the mouse onto the mouse pad without pressing either button, then click the left mouse button.
15 Open Drawers		Pull each **highlighted drawer** with its assigned hand past 60 percent travel, crossing both thresholds within 1.2 seconds.
16 Open Laptop[V]		Lift the laptop display from its front edge with the right hand and open it to 60 degrees.
17 Open Latched Door		Depress the near lever past 45 degrees with the left hand for five frames. Push the unlatched door to 45 degrees with the right hand, avoiding the table, frame, and far handle.
18 Open Micropipette Box		Open the hinged lid of the micropipette box until it is fully open.

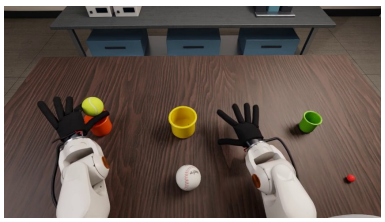
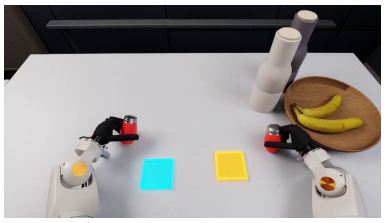
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Table 16 | Task profiles of DexOmni (continued).

Task	Screenshot	Task Instruction
19 Orient Cup Upright		Pick up the cup and place it upright on the table.
20 Pass and Orient Pen		Pass the pen from right-hand-only contact through both-hand contact to left-hand-only contact without dropping it. Hold it tip-up within 2 degrees of vertical.
21 Pass Block		Pick up the left block with the left hand, transfer it to the right hand in the middle, then place it stably inside the right box.
22 Pass Screwdriver		Touch the screwdriver with the right hand first, then touch it with the left hand while it is airborne, and let it land in the horizontal green target zone.
23 Photograph Logo[J]		Pick up the camera, aim it at the logo, and press the shutter button.
24 Place Ball in Bowl		Pick the **tennis ball** and place it into the bowl.
25 Place Cup on Plate		Pick up the mug and place it upright in the center of the plate.

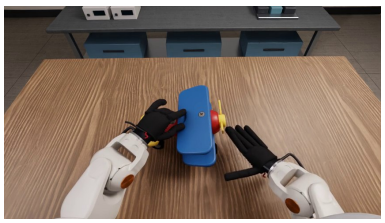
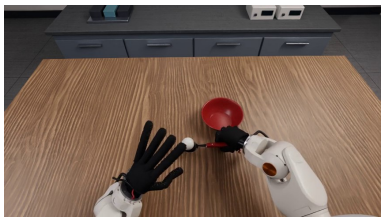
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Table 16 | Task profiles of DexOmni (continued).

Task	Screenshot	Task Instruction
26 Place Fruit in Bucket		Pick the **orange** and place it into the bucket.
27 Place Largest Fitting Ball		Choose the largest ball that fits the **large yellow J cup**. Place it in that cup, release both hands, and leave ball and cup stable.
28 Place Tableware		Pick up the **mug** and place it **above and to the right of the plate**, **upright with its opening facing up**. Move only the **mug** and leave the plate and distractors in place.
29 Place Wine Glass		Place the wine glass upright at the plate center, release it, and leave it stable.
30 Pour Seasonings		Use the assigned hands to pour salt into the cyan target and pepper into the orange target, with overlapping pouring. Keep each jar's total pouring time below four seconds.
31 Prepare Coffee		Take the mug from the cabinet and close both doors. Place it under the coffee outlet, press the green button, and brew for two seconds. Release the filled mug upright on the serving tray.
32 Press Soap Pump		Stabilize the sponge with the left hand, align the soap nozzle above it with the right hand, and physically depress the dispenser.

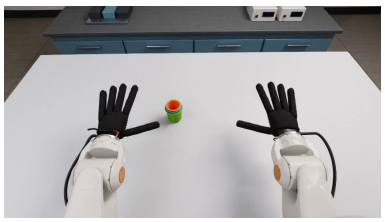
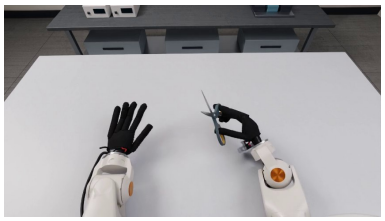
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Table 16 | Task profiles of DexOmni (continued).

Task	Screenshot	Task Instruction
33 Reposition Marker Grasp		Lift the capped marker with the right hand. Move both thumb and index contact points at least 2.5 cm toward its writing tip while keeping it lifted and grasped.
34 Roll Ball Through Gate		Roll or flick the tennis ball from its starting side through the opening between the cracker box and the sugar box. Keep it on the table without picking it up or touching either box.
35 Rotate Propeller		Spin the airplane propeller through three full turns of accumulated travel, in either direction.
36 Scoop Marbles		Keep the spoon level or slightly nose-down, scoop the marble, carry it over the bowl, and pour it into the bowl.
37 Scrub Bottle		Hold the water bottle with the left hand. Brush its lower, middle, then upper bands with the right hand. Release both objects, leaving the bottle upright and stable.
38 Set Dual Knobs		Turn both knobs together using their assigned hands, set the left knob to **60 degrees** counter-clockwise and the right knob to **60 degrees** clockwise, release both knobs, and keep the settings stable.
39 Set Timer		Hold the timer with the left hand. Enter **1 minute** and **1 second** with the right, press START last, then withdraw both hands for one countdown tick. Clear incorrect entries first.

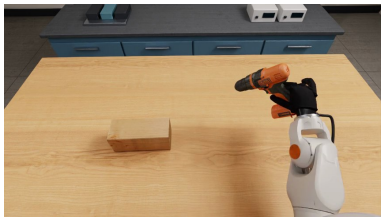
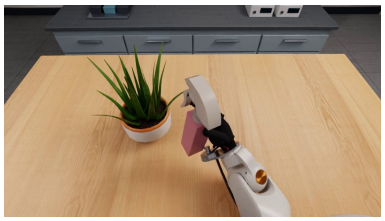
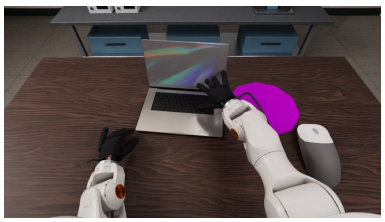
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Table 16 | Task profiles of DexOmni (continued).

Task	Screenshot	Task Instruction
40 Slide Knife Blade[V]		Pick up the utility knife with the right hand, then use the left fingertips on its slider to **extend** the blade fully. (The action goal is random: **extend** may be replaced with **retract**.)
41 Sort Nested Cups		Separate the three nested YCB cups, then place them upright in a straight row from left to right in the main camera: large, medium, small. Release all cups.
42 Sort Objects		Find six targets among the clutter and sort boxes, cans, and dishes into their marked baskets. Transfer the far-right target through both hands before placing it.
43 Squeeze Scissors		Lift the scissors clear of the table and make three complete cuts in the air. Try one hand first; the other hand may reposition or assist.
44 Stack Blocks		Arrange the four blocks into four distinct height layers with the large wood block on the bottom layer, then keep the hand from touching the blocks for 0.5 seconds.
45 Stir Pot		Hold the open pot with the left hand and stir with the right-hand spoon through the six ordered checkpoints and over 650 degrees.
46 Store Charger[VA]		Hold the drawer open with the left hand and place the charger fully inside with the right. Release the charger, then the drawer, and hold stable for half a second.

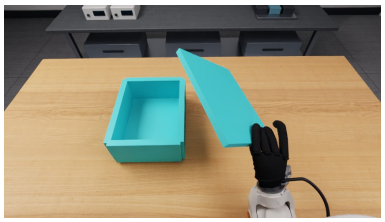
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Table 16 | Task profiles of DexOmni (continued).

Task	Screenshot	Task Instruction
47 Trigger Drill on Block		Use the right hand to bring the drill into contact with the wood block, then press the drill's trigger and hold it for half a second while keeping the contact.
48 Unfold Fan		Use both hands to grasp the two outermost fan ribs and unfold the fan symmetrically to **60 degrees**, then hold it steadily on the table.
49 Unfold Ruler		Pinch the two outer ruler segments with their assigned hands and unfold them together until the ruler is straight.
50 Water Plant[J]		Grasp the spray bottle and aim it at the plant before pressing, then press the sprayer so that the water lands on the plant or its soil.
EVAL-Comb01 Cover Pitcher and Box		First align and seat the pitcher lid, then release it and let it settle. Next align and seat the clear-box lid, release it and let it settle. Both lids must remain correctly seated on their own containers.
EVAL-Comb02 Open Laptop and Click Mouse		Open the initially closed laptop to the **60 degrees** with either hand and let it settle; you may release the screen. Move the mouse onto its pad without pressing either button, then click the left button once. Keep the laptop at the **target angle** and the mouse on the pad while the scene settles.
EVAL-Comb03 Pass Block and Close Box		Pick up the left block with the left hand, hand it to the right hand in the middle, and release it stably inside the clear box. Then use the right hand to pick up the lid, align and close the box, and release.

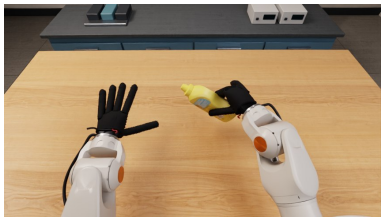
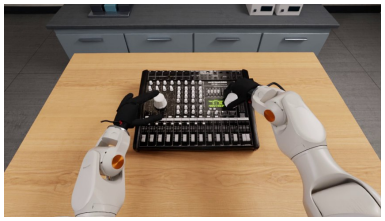
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Table 16 | Task profiles of DexOmni (continued).

Task	Screenshot	Task Instruction
EVAL-Comb04 Place Fruit and Photograph Logo		Pick the **orange**, place it in the bucket and release it. Then pick up the camera, aim at the logo on the bucket and press the shutter.
EVAL-Comb05 Scrub Bottle Then Fill Cup		Stabilize the bottle with the left hand and grasp the cleaning brush by its handle with the right hand. Scrub the bottle bands in order, either lower-middle-upper or upper-middle-lower, then release both objects and leave the bottle upright and stable. Next, hold the mug under the faucet with the left hand and open the faucet with the right hand. Keep the stream entering the upright mug for 0.5 seconds, then place the mug on the marked counter pad outside the sink and close the faucet. Leave the scrubbed bottle upright.
EVAL-Comb06 Season and Stir		Use the left hand for salt and the right hand for pepper. Pour both into the same open pot together, then put down and release the jars. Hold the pot with the left hand and stir with the right-hand spoon through all six angular sectors and more than 650 degrees, keeping the pot stable.
EVAL-Gen01 Cover Lunchbox		Align the lunchbox lid and place it onto the lunchbox.
EVAL-Gen02 Open Trash Bin		The trash bin is fixed after reset at a randomized tabletop pose and yaw. Open its hinged lid from closed to at least 75 degrees. Success is immediate when the physical lid reaches that opening. Either hand may be used; no assigned contact, release, or stable hold is required.

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Table 16 | Task profiles of DexOmni (continued).

Task	Screenshot	Task Instruction
EVAL-Gen03 Orient Bottle Upright		Pick up the soap bottle and place it upright on the table with its bottom facing down. Release the bottle and keep it upright for 0.5 seconds.
EVAL-Gen04 Pass Glue Stick		Pick up the glue stick with the left hand, transfer it to the right hand in the middle, then place it stably inside the right tray and release it.
EVAL-Gen05 Set Mixer Knobs		Use your left hand to turn the left knob counterclockwise and your right hand to turn the right knob clockwise, in either order. Set the left knob to **60 degrees** and the right knob to **-60 degrees**, release both knobs, and keep the settings stable.
EVAL-Gen06 Spray Shoe		Grasp and aim the spray bottle at the shoe upper before squeezing the trigger. Keep spraying the shoe to complete the task; a spray that entirely misses the shoe upper fails the attempt.
EVAL-Gen07 Stack Boxes		Arrange the four package boxes into four distinct height layers with the large package box on the bottom layer, then keep the hand from touching the boxes for 0.5 seconds.
EVAL-Flex01 Compress Sponge into Mug		Use the left hand to stabilize the YCB mug. Sponge 3 uses an isotropic soft-foam material, so clamp it from any reachable pair of opposing sides and compress it along X, Y, or Z; close slowly until it is compressed, keep the pinch while lifting it edge-on through the mug rim until it is mostly inside, then release both objects and leave the mug upright and stable.

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Table 16 | Task profiles of DexOmni (continued).

Task	Screenshot	Task Instruction
EVAL-Flex02 Fold Towel Twice		Fold the flat towel in half twice, in any directions. The two folds may be in the same direction or different directions. Release the folded towel and leave it stable on the table.
EVAL-Flex03 Pack Pillow		Use both hands to squeeze and lift the foam pillow, lower it into the open carton so it rests inside with most of the pillow below the rim, then release both hands and keep it stable for one second. Tilted placement and raised corners are allowed.